

Sensitivity Analysis in Sharp Regression Discontinuity Designs*

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Abstract

This paper develops a framework for extrapolating causal effects in sharp regression discontinuity (RD) designs beyond the local neighborhood of the cutoff. We adopt a partial identification approach that yields bounds on a broad class of causal parameters, including average and distributional treatment effects. Identification relies on smoothness restrictions on the distribution of potential outcomes with respect to the running variable, parameterized by sensitivity measures that capture the degree of smoothness and admit an interpretation in terms of statistical dependence. Within this framework, we derive sharp bounds on counterfactual distributions that can be estimated using standard nonparametric, semiparametric, or

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flexible parametric methods. The approach is well suited for sensitivity analysis, providing a transparent assessment of the robustness of extrapolated RD estimates. Monte Carlo simulations demonstrate its finite-sample performance. We also present an empirical application to illustrate how the proposed approach can be used to address empirical questions concerning external validity in sharp RD designs.

Keywords: regression discontinuity, sharp designs, extrapolation, external validity, partial identification, sensitivity analysis.

JEL Classification Numbers: C14, C31, C36, C51.

1 Introduction

Regression discontinuity (RD) designs are among the most widely used methods for causal inference with observational data. They apply to settings in which treatment assignment is determined by whether an observed score (i.e., the running variable) crosses a known cutoff value. Exploiting this feature, RD estimates are obtained by comparing the average outcome for observations located just above the cutoff with that for observations located just below the cutoff. Since their introduction by [Thistlethwaite and Campbell \(1960\)](#), RD designs have been increasingly used in the economics literature to estimate causal treatment effects.¹

The seminal paper by [Hahn et al. \(2001\)](#) establishes conditions under which the RD estimate admits a causal meaning. The key condition is that the conditional expectation functions of the potential outcomes on the running variable are continuous at the cutoff value, which is referred to as “continuity assumption.” [Hahn et al. \(2001\)](#) show that under the continuity assumption, the RD estimate is the average treatment effect (ATE) at the cutoff level in sharp RD designs when the treatment effects are heterogeneous. This identification result gives rise to an empirical challenge in that regression discontinuity estimates recover only local causal effects. The term “local” refers to treatment effects for a specific subpopulation—namely, units whose running variable values lie in a neighborhood of the cutoff—rather than effects for the entire population. Consequently, it is difficult to assess the external validity of regression discontinuity estimates, even though several empirical studies are explicitly concerned with this issue (e.g., [Abdulkadiroğlu et al. \(2014\)](#); [Dobbie and Fryer Jr \(2014\)](#); [Hainmueller et al. \(2015\)](#)).

This paper develops a framework for extrapolating causal effects in sharp regression discontinuity designs. The proposed approach enables identification of treatment effects for broader

¹[Lee and Lemieux \(2010\)](#) and [Cattaneo and Titiunik \(2022\)](#) provide comprehensive surveys of RD designs.

subpopulations—and, under additional restrictions, potentially for the entire population—beyond the set of units at the cutoff. To achieve this, we adopt a partial identification strategy that delivers bounds on the causal parameters of interest. The partial identification approach allows us to learn about a broader class of causal parameters in sharp RD designs under weaker identifying assumptions, thereby yielding results that are more robust to deviations from point-identifying restrictions and improving the credibility of the analysis. As a byproduct, the framework naturally lends itself to sensitivity analysis.

Our key identifying assumption is a smoothness restriction. Specifically, we parameterize the degree of smoothness of the conditional distribution functions of the potential outcomes on the running variable. These parameters also serve as a sensitivity measure with respect to the identifying assumption. Importantly, they admit a natural interpretation in terms of the degree of statistical dependence between the potential outcomes and the running variable. These parameters are selected and varied by the researcher, thereby facilitating sensitivity analysis. Note that our identifying assumption is closely related to those introduced in [Kline and Santos \(2013\)](#), [Kim et al. \(2018\)](#), and [Masten and Poirier \(2018\)](#).

We derive the *sharp bounds* on the counterfactual distributions of the potential outcomes. These sharp bounds are tractable and easy to estimate in practice. Notably, the sharp bounds on the counterfactual distributions allow us to construct sharp bounds on functionals of these distributions, which serve as building blocks for identification of causal parameters of interest, such as the ATE and quantile treatment effects (QTEs), for broad subpopulations. In doing so, our framework can address the issue about external validity of RD estimates.

Since the resulting bounds may be wide, we introduce an additional identifying assumption that can help tighten them. Motivated by the stochastically monotone instrumental variable assumption of [Mourifie et al. \(2020\)](#), we propose the *stochastically monotone running variable assumption*. In an appropriate sense, this assumption characterizes the direction in which the conditional distribution functions of the potential outcomes change with the running variable as the conditioning value increases. As a consequence, it narrows the bounds on the counterfactual distributions of the potential outcomes and thereby yields tighter bounds on the causal parameters.

Based on the identification results, we conduct a small Monte Carlo simulation study and perform a sensitivity analysis. The simulation results demonstrate that our framework is effective for extrapolating regression discontinuity estimates, thereby allowing us to learn about a broader class of causal parameters in sharp RD designs.

We revisit the empirical study of [Pop-Eleches and Urquiola \(2013\)](#), who examine the effect of

attending better schools on students’ academic performance, and use it to illustrate the empirical relevance of our approach. Our findings indicate that, with respect to the QTE, the RD estimates may not be representative of the relevant causal parameters. Specifically, over a wide range of quantile levels, the estimated bounds on the QTE conditional on values of the running variable away from the cutoff do not contain the RD estimates obtained at the cutoff. In addition, the estimated bounds on the unconditional QTE do not necessarily contain the RD estimates of the QTE at the cutoff. These findings suggest that, abstracting from sampling error, the QTE may vary across subpopulations indexed by the running variable, implying that the RD estimates at the cutoff need not be representative of the causal effect either for other subpopulations or for the population as a whole. In contrast, we do not find comparably clear evidence of such heterogeneity for the ATE.

We contribute to the literature in several ways. First, we develop a framework that enables extrapolation of regression discontinuity estimates to a broad class of causal parameters in sharp RD designs. By moving beyond treatment effects local to the cutoff, our framework directly addresses questions of external validity and enhances the policy relevance of RD estimates. In addition, we derive the *sharp* bounds on the conditional distributions of potential outcomes with respect to the running variable, which in turn enable construction of sharp bounds on distributional treatment effects in sharp regression discontinuity designs.

A growing body of literature has examined related concerns, and our approach contributes to this line of research by providing a partial identification approach. Angrist and Rokkanen (2015) propose an approach to extrapolation of RD estimates that relies on a conditional independence assumption.² Their approach requires the existence of a set of covariates such that the running variable is conditionally independent of the potential outcomes given these covariates. Cattaneo et al. (2021) develop a method for extrapolating regression discontinuity estimates based on multiple cutoffs, which shares conceptual similarities with difference-in-differences designs. Deaner and Kwon (2025) exploit a comonotonicity assumption to extrapolate the conditional average treatment effect away from the cutoff, imposing conditional monotonicity on the conditional expectation functions of the potential outcomes with respect to the running variable. Feng and Ruggieri (2025) propose a structural modeling framework for extrapolating average partial effects in regression discontinuity designs. Our approach does not require additional variables or structural modeling and imposes no distributional assumptions. As a result, we complement these studies by adopting a partial

²Rokkanen (2015) also considers a similar approach to extrapolation of RD estimates in sharp designs.

identification strategy based on a parsimonious modeling framework.³

Second, we derive identification results for a broader class of causal parameters than those considered in prior studies. All of the aforementioned studies have focused primarily on the extrapolation of the average RD treatment effect. In contrast, we provide results on the extrapolation of not only the average RD treatment effect but also distributional RD treatment effects. Consequently, our results allow researchers to investigate heterogeneity in treatment effects, in addition to the extrapolation of RD treatment effects. Sun (2023) adopts a partial identification approach under an assumption similar to ours, but her analysis focuses primarily on the extrapolation of average treatment effects in RD designs with potentially multiple cutoff values.

Third, this paper contributes to the literature on sensitivity analysis. Sensitivity analysis is valuable for empirical researchers seeking to assess the robustness of empirical conclusions to variations in identifying assumptions, data, or modeling choices. Our framework builds primarily on the approaches developed by Kline and Santos (2013) and Masten and Poirier (2018), who propose sensitivity analysis frameworks in their respective settings.⁴

The rest of this paper is organized as follows. In Section 2, we introduce the model and identifying assumptions. Section 3 provides the identification results and formalizes the idea of the stochastically monotone running variable assumption. Section 4 briefly discusses the estimation of bounds, and section 5 presents the Monte Carlo simulation results. Section 6 provides an empirical illustration, and Section 7 concludes. All mathematical proofs are provided in Appendix.

2 Model and Assumptions

Let $(\Omega, \mathcal{S}, \mathbf{P})$ be a probability space. Let R denote the running variable and c be the cutoff level. Then, in a sharp RD design, $D = \mathbf{1}(R \geq c)$, where D is the observed binary treatment variable and $\mathbf{1}(\cdot)$ is an indicator function. Letting Y_1 and Y_0 be the potential outcomes, the observed outcome variable Y is $Y = DY_1 + (1 - D)Y_0$.

Let h be a measurable function such that $E[|h(Y_d)|] < \infty$ for all $d \in \{0, 1\}$. For a generic set A , $\bar{A}$ is the closure of A . We denote the support of a random variable W by $Supp(W)$ and the distribution function by $F_W(\cdot)$.⁵ We define the “treated region” and “control region” as $\mathcal{R}_1 \equiv [c, \infty) \cap Supp(R)$

³Although we focus on extrapolation in sharp RD designs, it is worth noting that a number of papers in the literature address extrapolation in fuzzy RD designs, including Battistin and Rettore (2008), Mealli and Rampichini (2012), and Bertanha and Imbens (2020).

⁴There are a growing number of papers that consider sensitivity analysis, including Kline and Santos (2013), Manski and Pepper (2018), Masten and Poirier (2018), Masten and Poirier (2020), Rambachan and Roth (2023), and Masten et al. (2024), among others.

⁵By definition, $Supp(W)$ is the smallest closed set outside of which $F_W(\cdot)$ is either 0 or 1 (cf. Folland (1999, p.132)).

and $\mathcal{R}_0 \equiv (-\infty, c) \cap \text{Supp}(R)$, respectively.

Assumption 2.1. *The sets $\mathcal{R}_1$ and $\mathcal{R}_0$ are nonempty, and for any $r_1 \in \mathcal{R}_1$ and $r_0 \in \mathcal{R}_0$,*

$$E[h(Y_1) | R = r_1] = E[h(Y) | R = r_1]$$

and

$$E[h(Y_0) | R = r_0] = E[h(Y) | R = r_0].$$

Assumption 2.1 requires that $E[h(Y_1) | R_1 = r_1]$ and $E[h(Y_0) | R = r_0]$ are identified from the distribution of (Y, D, R) and provides baseline estimators for extrapolation of treatment effects. Specifically, for given $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$, we use $E[h(Y_1) | R = r_1]$ to learn about $E[h(Y_1) | R = r_0]$ and use $E[h(Y_0) | R = r_0]$ to learn about $E[h(Y_0) | R = r_1]$. When $c \in \text{Supp}(R)$, the treatment effect at the cutoff can be identified under continuity assumptions (cf. [Hahn et al. \(2001\)](#)). It is also worth pointing out that $r_1 \geq r_0$ for any $r_d \in \mathcal{R}_d$ with $d \in \{0, 1\}$.

The next assumption is our main assumption for the extrapolation of treatment effects in sharp RD designs.

Assumption 2.2. *There exist $k_{0,h}, k_{1,h} \geq 0$ such that for all $r, \tilde{r} \in \text{Supp}(R)$,*

$$\left| E[h(Y_1) | R = r] - E[h(Y_1) | R = \tilde{r}] \right| \leq k_{1,h} |r - \tilde{r}|$$

and

$$\left| E[h(Y_0) | R = r] - E[h(Y_0) | R = \tilde{r}] \right| \leq k_{0,h} |r - \tilde{r}|.$$

Assumption 2.2 is a smoothness condition imposed on the conditional expectation functions. When $c \in \text{Supp}(R)$ and $h(y) = y$, it is stronger than the continuity-at-the-cutoff condition in [Hahn et al. \(2001\)](#) (i.e., Assumption (A1) in [Hahn et al. \(2001\)](#)). While Assumption (A1) in [Hahn et al. \(2001\)](#) only requires continuity at $R = c$, Assumption 2.2 requires $E[h(Y_d) | R = r]$ to be globally Lipschitz continuous in r for $d \in \{0, 1\}$.

The constants $k_{0,h}$ and $k_{1,h}$ can be interpreted as sensitivity parameters for smoothness, as in [Kim et al. \(2018\)](#). They may also be interpreted as capturing the degree of mean independence between $h(Y_d)$ and R . Given the generality of h , these constants can further be understood as measuring the degree of independence between the potential outcomes and the running variable. In particular, when $k_{d,h} = 0$ for a given $d \in \{0, 1\}$ and $h(Y) = h_y(Y) \equiv \mathbf{1}(Y \leq y)$ for $y \in \mathbb{R}$,

Assumption 2.2 implies that Y_d is independent of R , as in Kline and Santos (2013) and Masten and Poirier (2018). Note that these constants are chosen by researchers.

There are several papers in the literature that consider restrictions similar to those in Assumption 2.2. For example, Kim et al. (2018) introduce smooth treatment response and smooth monotone treatment response assumptions of Lipschitz continuity of the conditional expectation functions of the potential outcomes with respect to the treatment. In addition, Manski and Pepper (2018) consider similar assumptions that impose bounded variation on potential outcomes with respect to treatments. Assumption 2.2 differs from them in that the smoothness assumption is imposed with respect to the conditioning variable, namely the running variable.⁶

3 Identification

The following theorem provides bounds on the conditional expectation functions that will serve as the building blocks for constructing sharp identified regions for a certain causal parameter of interest.

Theorem 3.1. *Suppose that Assumptions 2.1 and 2.2 hold. Let h be a measurable function such that $E[|h(Y_d)|] < \infty$ for all $d \in \{0, 1\}$. Then, for any $r \in \text{Supp}(R)$, we have*

$$\begin{aligned} E[h(Y_1) | R = r] &\in [LB_1(k_{1,h}, r), UB_1(k_{1,h}, r)], \\ E[h(Y_0) | R = r] &\in [LB_0(k_{0,h}, r), UB_0(k_{0,h}, r)], \end{aligned}$$

where

$$\begin{aligned} LB_1(k_{1,h}, r) &\equiv \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[h(Y_1) | R = \tilde{r}] - k_{1,h} \cdot |r - \tilde{r}| \right\}, \\ UB_1(k_{1,h}, r) &\equiv \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[h(Y_1) | R = \tilde{r}] + k_{1,h} \cdot |r - \tilde{r}| \right\}, \\ LB_0(k_{0,h}, r) &\equiv \sup_{\tilde{r} \in \mathcal{R}_0} \left\{ E[h(Y_0) | R = \tilde{r}] - k_{0,h} \cdot |\tilde{r} - r| \right\}, \\ UB_0(k_{0,h}, r) &\equiv \inf_{\tilde{r} \in \mathcal{R}_0} \left\{ E[h(Y_0) | R = \tilde{r}] + k_{0,h} \cdot |\tilde{r} - r| \right\}. \end{aligned}$$

⁶A related assumption in the literature is the monotone treatment selection (MTS) assumption introduced by Manski and Pepper (2000). However, the MTS assumption imposes monotonicity on the conditional expectation of the potential outcome with respect to the counterfactual treatment variable, which differs from Assumption 2.2. In addition, the monotone instrumental variables (MIV) assumption in Manski and Pepper (2000) is also closely related to Assumption 2.2. Nevertheless, unlike MTS or MIV, we do not impose monotonicity on the conditional expectation functions. Instead, we restrict the range within which the counterfactual conditional expectation functions can lie.

The bounds in Theorem 3.1 have a similar structure to that of Manski’s bounds under the monotone instrumental variables bounds (cf. Manski and Pepper (2000)) and bounds on the counterfactual mean function in Kim et al. (2018).

We can construct bounds on the conditional expectation functions over regions in which they may not be point identified. Suppose, for example, that $\text{Supp}(R) = [r_{\min}, r_{\max}]$ for some constants $r_{\min}$ and $r_{\max}$ with $r_{\min} < c < r_{\max}$ and that $\mathcal{R}_1 = [c, r_{\max}]$ and $\mathcal{R}_0 = [r_{\min}, c)$. Focusing on the case that $h(y) = y$, it is evident that the counterfactual mean $E[Y_d|R = r]$ is not identified for $r \in \mathcal{R}_{1-d}$ without further restrictions. Nevertheless, we can partially identify $E[Y_d|R = r]$ for $r \in \mathcal{R}_{1-d}$ by using the results in Theorem 3.1.

The bounds can be constructed by replacing the conditional expectation functions with their consistent estimators—whether parametric, semiparametric, or nonparametric—and then taking the supremum and/or infimum over the relevant region. However, in sharp RD designs, these intersection bounds in Theorem 3.1 can be further reduced. To this end, let $r_1^* \equiv \inf \mathcal{R}_1$ and $r_0^* \equiv \sup \mathcal{R}_0$.

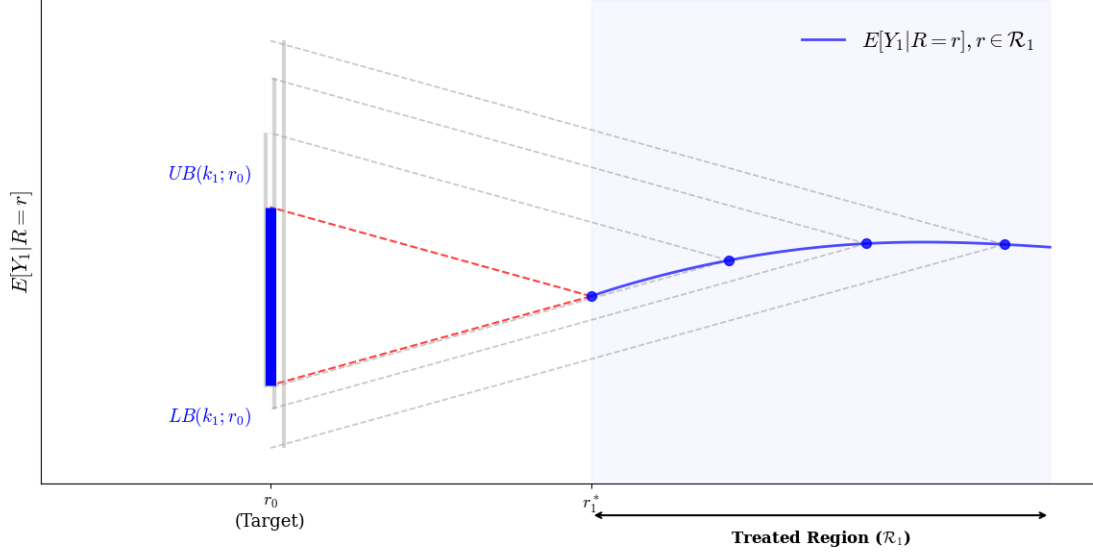
Theorem 3.2. *Suppose that Assumptions 2.1 and 2.2 hold. Let $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$. Then, we have*

$$\begin{aligned} LB_1(k_{1,h}, r_0) &= E[h(Y_1)|R = r_1^*] - k_{1,h} \cdot |r_0 - r_1^*|, \\ UB_1(k_{1,h}, r_0) &= E[h(Y_1)|R = r_1^*] + k_{1,h} \cdot |r_0 - r_1^*|, \\ LB_0(k_{0,h}, r_1) &= E[h(Y_0)|R = r_0^*] - k_{0,h} \cdot |r_1 - r_0^*|, \\ UB_0(k_{0,h}, r_1) &= E[h(Y_0)|R = r_0^*] + k_{0,h} \cdot |r_1 - r_0^*|. \end{aligned}$$

Theorem 3.2 allows us to avoid searching over $\mathcal{R}_1$ and $\mathcal{R}_0$ to compute the supremum and infimum in the intersection bounds in Theorem 3.1. the argsup and arginf to construct the bounds. To illustrate the empirical relevance of the results, suppose that $\text{Supp}(R) = [r_{\min}, r_{\max}]$ for some constants $r_{\min}$ and $r_{\max}$ with $r_{\min} < c < r_{\max}$ and that $\mathcal{R}_1 = [c, r_{\max}]$ and $\mathcal{R}_0 = [r_{\min}, c)$, and therefore, $r_1^* = r_0^* = c$. In this case, Theorem 3.2 implies that estimating the bounds requires only the estimation of $E[h(Y_1)|R = c] = E[h(Y)|R = c]$ and $E[h(Y_0)|R = c] = \lim_{r \uparrow c} E[h(Y)|R = r]$, which can be implemented using the well-known nonparametric approach proposed by Calonico et al. (2014).

The intuition behind the results in Theorem 3.2 is based on the geometric fact that, for a fixed $r_0 \in \mathcal{R}_0$ and for any $\tilde{r} \in \mathcal{R}_1$, r_1^* is the unique point such that $|r_0 - \tilde{r}| = |r_0 - r_1^*| + |\tilde{r} - r_1^*|$. Moving

Figure 3.1: Extrapolation at the Boundary



from r_1^* to an interior point of $\mathcal{R}_1$ increases the distance to r_0 by $|\tilde{r} - r_1^*|$. Under Assumption 2.2, the maximal admissible change in the value of $g_1(\tilde{r})$ from $g_1(r_1^*)$ is $k_{1,h}|\tilde{r} - r_1^*|$.

Figure 3.1 illustrates this intuition. The blue solid line in Figure 3.1 represents $g_1(r) = E[Y_1|R=r]$ over $\mathcal{R}_1$ (i.e., $h(x) = x$). Our objective is to identify $E[Y_1|R=r_0]$ for a given $r_0 \in \mathcal{R}_0$, which is the target point. The dashed lines correspond to the upper and lower bounds on $E[Y_1|R=r_0]$, constructed by using $E[Y_1|R=r_1]$ for $r_1 \in \mathcal{R}_1$, which are indicated by the circles on the blue solid line. As r_1 moves farther away from r_0 , the bounds widen as the $k_{1,h}|r_1 - r_0|$ increases. Focusing on the lower bound, recall that under Assumption 2.2, for a fixed $r_0 \in \mathcal{R}_0$,

$$E[Y_1|R=r_0] \geq E[Y_1|R=r_1] - k_{1,h} \cdot |r_0 - r_1|$$

for $r_1 \in \mathcal{R}_1$. Even when $E[Y_1|R=r]$ is assumed to be increasing in r , the potential gain from choosing a larger r_1 , which would make $E[Y_1|R=r_1]$ closer to $E[Y_1|R=r_0]$, is dominated by the term $k_{1,h} \cdot |r_0 - r_1|$ under Assumption 2.2. Consequently, the bound is tightened only by selecting $r_1 \in \mathcal{R}_1$ as close as possible to r_0 . Owing to the sharp RD feature, the closet point to r_0 in $\mathcal{R}_1$ is r_1^* .

It is worth pointing out that $r_d^* \in \mathcal{R}_d$ is not required for the results in Theorem 3.2 to hold and it is enough to assume that $r_d^* \in \text{Supp}(R)$. If $r_d^* \notin \mathcal{R}_d$, $E[h(Y_d)|R=r_d^*]$ can be understood as the

one-side limit of the corresponding conditional mean function, $\lim_{r \rightarrow r_d^*: r \in \mathcal{R}_d} E[h(Y_d) | R = r]$. In addition, Theorem 3.2 facilitates the proof of sharpness for some specific choice of h .⁷

3.1 Identification of the Distributions of Counterfactual Outcomes

We now turn to the identification of the distributions of counterfactual outcomes. Identification of distribution functions is important because it enables the analysis of distributional treatment effects, such as quantile treatment effects, and facilitates the study of heterogeneous treatment effects across the distribution of the outcome variable of interest. Several papers in the literature examine distributional effects in regression discontinuity designs, including Frandsen et al. (2012), Shen and Zhang (2016), and Qu and Yoon (2019) just to name a few. Building on this literature, we extend our identification results for conditional mean functions to the distributions of potential outcomes. For generic random variables U and V , we denote the conditional distribution of U on $V = v$ by $F_{U|V}(\cdot|v)$.

Assumption 3.1. *The following conditions hold:*

- (i) *For each $d \in \{0, 1\}$ and for all $y \in \mathbb{R}$, $F_{Y_d|R}(y|r_d)$ is identified for any $r_d \in \mathcal{R}_d$.*
- (ii) *There exist $k_{0,c}, k_{1,c} \geq 0$ such that for all $r, \tilde{r} \in \text{Supp}(R)$,*

$$\begin{aligned} \sup_{y \in \mathbb{R}} \left| F_{Y_1|R}(y|r) - F_{Y_1|R}(y|\tilde{r}) \right| &\leq k_{1,c} |r - \tilde{r}|, \\ \sup_{y \in \mathbb{R}} \left| F_{Y_0|R}(y|r) - F_{Y_0|R}(y|\tilde{r}) \right| &\leq k_{0,c} |r - \tilde{r}|. \end{aligned} \tag{1}$$

Assumption 3.1-(i) requires that we can identify the conditional distributions of counterfactual outcomes for some subpopulations. Frandsen et al. (2012) and Shen and Zhang (2016) consider the identification of conditional distributions of the potential outcomes on the running variable at the cutoff point. Similar to Hahn et al. (2001), the key identifying assumption is the continuity assumption at the cutoff value. Specifically, they require that for all $y \in \mathbb{R}$ and $d \in \{0, 1\}$, $F_{Y_d|R}(y|r)$ is continuous at $r = c$. Under this assumption, they show that distributional causal parameters can be identified.

Motivated by Kline and Santos (2013) and Masten and Poirier (2018), Assumption 3.1-(ii) imposes a condition that the distance between two distributions is small when the conditioning values are close. Specifically, the left hand sides of the equations in Assumption 3.1-(ii) are the

⁷See the proof of Theorem 3.3.

Kolmogorov-Smirnov distance between two distributions, whereas the right hand sides are distances between r and $\tilde{r}$. Related to this point, Assumption 3.1-(ii) can be interpreted as the continuity assumption on the distribution functions, which strengthens the continuity assumption in Frandsen et al. (2012) and Shen and Zhang (2016) in the sense that the distribution functions are continuous on $Supp(R)$.

Moreover, Assumption 3.1-(ii) characterizes the statistical dependence between the potential outcomes and running variable. When $k_{d,c} = 0$, the potential outcome Y_d is statistically independent of the running variable R . As the value of $k_{d,c}$ increases, a greater degree of dependence between Y_d and MR is allowed. Consistent with this interpretation, a number of studies in the literature adopt assumptions that are similar in spirit to Assumption 3.1-(ii). Kline and Santos (2013) consider a restriction on the distance between the distributions of the outcome variable for selected and non-selected units, which captures the degree to which a missing-at-random assumption holds in a sample selection framework. Masten and Poirier (2018) introduce the concept of c -independence, which parameterizes the degree of conditional independence in a potential outcomes model with a binary treatment. Although Rambachan and Roth (2023) do not directly characterize the degree of dependence between variables, they propose a related identifying assumption that quantifies the degree of violation of the parallel trends assumption in a difference-in-differences framework through a nuisance parameter analogous to $k_{d,c}$ in this paper.

It is important to note that the choice of $k_{d,c}$ for $d \in \{0, 1\}$ can affect the identifying power of Assumption 3.1-(ii). Since the left hand sides in (1) are bounded by 1, there is no restriction imposed on the distributions when the right hand sides are greater than or equal to 1. Related to this point, we point out that it is possible to rescale the running variable before choosing the value of $k_{d,c}$. Specifically, for any affine transformation of R , $R' = aR + b$ with $a \neq 0$ and $b \in \mathbb{R}$, Assumption 3.1-(ii) can be rewritten as $\sup_{y \in \mathbb{R}} |F_{Y_d|R'}(y | r') - F_{Y_d|R'}(y | \tilde{r}')| \leq k'_{d,c} |r' - \tilde{r}'|$ with $k'_{d,c} = k_{d,c}/|a|$, while the right hand side is $k'_{d,c} |r' - \tilde{r}'| = \frac{k_{d,c}}{|a|} |a| |r - \tilde{r}| = k_{d,c} |r - \tilde{r}|$. This implies that the condition $k_{d,c} |r - \tilde{r}| < 1$ is invariant to an affine transformation of R with the corresponding rescaling of $k_{d,c}$.

Theorem 3.3. *Suppose that Assumption 3.1 holds. Let $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$ be given and $d' \equiv 1 - d$. Define*

$$\begin{aligned} \underline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}) &\equiv \max \{ F_{Y_d|R}(y | r_d^*) - k_{d,c} |r_{d'} - r_d^*|, 0 \}, \\ \overline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}) &\equiv \min \{ F_{Y_d|R}(y | r_d^*) + k_{d,c} |r_{d'} - r_d^*|, 1 \}. \end{aligned} \tag{2}$$

Then, we have

(a) $F_{Y_d|R}(\cdot | r_{d'}) \in \mathcal{F}_{Y_d|r_{d'}}^{k_{d,c}}$, where

$$\mathcal{F}_{Y_d|r_{d'}}^{k_{d,c}} \equiv \left\{ F \in \mathcal{F}_{\mathbb{R}} : \underline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}) \leq F(y) \leq \overline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}), \forall y \in \mathbb{R} \right\};$$

(b) For each $d \in \{0, 1\}$ and for each $\varepsilon_d \in [0, 1]$ and $\eta_d \in (0, \min\{\delta_d, (1 - \delta_d)/2\})$, there exists a cdf $F_{Y_d|R}^{k_d}(\cdot | r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}_{Y_d|r_{d'}}^{k_d}$ such that

(i) for any $r_0 \in \mathcal{R}_0$ and $r_1 \in \mathcal{R}_1$, there exists a joint distribution of (Y_1, Y_0, R, D) consistent with our maintained assumptions such that, for all $y \in \mathbb{R}$,

$$\Pr(Y_1 \leq y | R = r_0) = F_{Y_1|R}^{k_{1,c}}(y | r_0; \varepsilon_1, \eta_1),$$

$$\Pr(Y_0 \leq y | R = r_1) = F_{Y_0|R}^{k_{0,c}}(y | r_1; \varepsilon_0, \eta_0);$$

(ii) for each $d \in \{0, 1\}$ and each $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y | r_d^*) < 1$, as $\eta_d \searrow 0$,

$$F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; 1, \eta_d) \nearrow \overline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'})$$

and

$$F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; 0, \eta_d) \searrow \underline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'});$$

(iii) for each $y \in \mathbb{R}$ and $r_{d'} \in \mathcal{R}_{d'}$, the function $\varepsilon_d \mapsto F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d)$ is continuous on $[0, 1]$;

(c) Consequently, for any $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y | r_d^*) < 1$, the pointwise bounds

$$F_{Y_d|R}(y | r_{d'}) \in \left[\underline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}), \overline{F}_{Y_d|R}^{k_{d,c}}(y | r_{d'}) \right]$$

have a sharp interior.

Theorem 3.3 establishes the sharp bounds on the conditional distribution functions of the potential outcomes on the running variable, which are useful for extrapolation of treatment effects. The proof of Theorem 3.3 builds on the proof of Proposition 2 in Masten and Poirier (2018), but there is a substantial difference between the two. Importantly, while Masten and Poirier (2018) consider the c -independence assumption in a binary treatment effect model—where the conditioning variable is binary—our results generalize their framework to settings in which the conditioning variable

(i.e., the running variable) may be either discrete or continuous. As a result, we cannot directly apply the results of [Masten and Poirier \(2018\)](#), since this generalization renders the establishment of sharpness of the bounds more technically challenging.

The bounds in (2) are sharp in the sense that, for $d \in \{0, 1\}$ and $d' = 1 - d$, there exist attainable distribution functions $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \epsilon_d, \eta_d)$ that can be made arbitrarily close to the lower and upper bounds and that vary continuously between these bound functions, as pointed out by [Masten and Poirier \(2018\)](#). These properties of the bounds are essential for constructing sharp identified regions for functionals of the conditional distribution functions that correspond to causal parameters of interest, such as (conditional) average or quantile treatment effects.

The identification result presented in Theorem (3.3) allows us to construct bounds on the unconditional distributions of the potential outcomes. Specifically, since the distribution of R , F_R , is identified from the data, it is possible to derive bounds on the unconditional distribution function of Y_d by using the fact that

$$F_{Y_d}(y) = \int F_{Y_d|R}(y|r) dF_R(r).$$

The following corollary provides the identification result for the unconditional distribution functions of the potential outcomes.

Corollary 3.4. *Suppose that Assumptions 3.1 and 3.2 hold. Let $d \in \{0, 1\}$ be given and $d' = 1 - d$. Then, for each $y \in \mathbb{R}$,*

$$F_{Y_d}(y) \in \left[\int_{\mathcal{R}_{d'}} \underline{F}_{Y_d|R}^{k_{d,c}}(y | r) dF_R(r) + \int_{\mathcal{R}_d} F_{Y|R}(y|r) dF_R(r), \int_{\mathcal{R}_{d'}} \overline{F}_{Y_d|R}^{k_{d,c}}(y | r) dF_R(r) + \int_{\mathcal{R}_d} F_{Y|R}(y|r) dF_R(r) \right].$$

3.2 Identification of Causal Parameters

We consider the identification of causal parameters, such as the conditional quantile treatment effect (CQTE) and conditional average treatment effect (CATE).

For each $d \in \{0, 1\}$ and $\tau \in (0, 1)$, define

$$Q_{Y_d|R}(\tau | r) \equiv \inf \{y \in \text{Supp}(Y_d | R = r) : F_{Y_d|R}(y | r) \geq \tau\}$$

as the τ -th conditional quantile function of the potential outcome Y_d on $R = r$. Then the τ -th CQTE given $R = r$ is

$$CQTE(\tau|r) \equiv Q_{Y_1|R}(\tau|r) - Q_{Y_0|R}(\tau|r).$$

From the CQTE, we can identify the CATE on $R = r$ as

$$CATE(r) \equiv E[Y_1 - Y_0 | R = r] = \int_0^1 CQTE(\tau | r) d\tau.$$

We provide closed-form bounds on the quantile function of the potential outcome Y_d given $R = r_{d'}$, where for each $d \in \{0, 1\}$, $d' = 1 - d$ and $r_{d'} \in \mathcal{R}_{d'}$. Let

$$\delta_d \equiv \delta_d(r_{d'}) \equiv k_{d,c} |r_{d'} - r_d^*|.$$

For $\tau \in (0, 1)$ and $r_{d'} \in \mathcal{R}_{d'}$, define

$$\begin{aligned} \underline{Q}_{Y_d|R}^{k_{d,c}}(\tau | r_{d'}) &:= \begin{cases} -\infty & \tau \leq \delta_d, \\ Q_{Y|R,D=d}(\tau - \delta_d | r_d^*) & \tau > \delta_d, \end{cases} \\ \overline{Q}_{Y_d|R}^{k_{d,c}}(\tau | r_{d'}) &:= \begin{cases} Q_{Y|R,D=d}(\tau + \delta_d | r_d^*) & \tau + \delta_d \leq 1, \\ +\infty & \tau + \delta_d > 1. \end{cases} \end{aligned}$$

For the identification of the quantile functions of the potential outcomes, we impose the following condition.

Assumption 3.2. *For each $d \in \{0, 1\}$, the conditional cdfs at the boundary point r_d^* , $F_{Y_d|R}(\cdot | r_d^*)$, are continuous and strictly increasing.*

Assumption 3.2 requires that the potential outcomes be continuous random variables. We now establish the identification of the CQTE.

Theorem 3.5. *Let $r \in \text{Supp}(R)$. Suppose that Assumptions 3.1 and 3.2 hold. For a given $\tau \in (0, 1)$, the $CQTE(\tau | r)$ lies in the set*

$$\begin{aligned} & \left[\underline{CQTE}^{k_c}(\tau | r), \overline{CQTE}^{k_c}(\tau | r) \right] \\ \equiv & \begin{cases} \left[Q_{Y|R,D=1}(\tau | r) - \overline{Q}_{Y_0|R}^{k_{0,c}}(\tau | r), Q_{Y|R,D=1}(\tau | r) - \underline{Q}_{Y_0|R}^{k_{0,c}}(\tau | r) \right] & \text{for } r \in \mathcal{R}_1, \\ \left[\underline{Q}_{Y_1|R,D=1}^{k_{1,c}}(\tau | r) - Q_{Y_0|R}(\tau | r), \overline{Q}_{Y_1|R}^{k_{1,c}}(\tau | r) - Q_{Y|R,D=0}(\tau | r) \right] & \text{for } r \in \mathcal{R}_0, \end{cases} \end{aligned} \quad (3)$$

where $k_c = (k_{0,c}, k_{1,c})$.

Moreover, the interior of this set is sharp.

The main idea of Theorem 3.5 is to invert the bounds on the counterfactual distribution, which is not identified over a certain region of the support of R . Since any quantile is a D_1 -parameter in the sense of Stoye (2010) (or D -parameter in the sense of Manski (1997)), the lower (respectively, upper) bound on the conditional quantile is obtained by inverting the upper (respectively, lower) bound on the conditional distribution of the corresponding potential outcome.

The bounds on the CQTE yield the sharp bounds on the CATE, and this is established in the following corollary:

Corollary 3.6. *Let $r \in \text{Supp}(R)$. Suppose that Assumptions 3.1 and 3.2 hold. Then, the $CATE(r)$ lies in the set*

$$\left[\int_0^1 \underline{CQTE}^{k_c}(\tau | r) d\tau, \int_0^1 \overline{CQTE}^{k_c}(\tau | r) d\tau \right].$$

Moreover, this identified set is sharp.

The unconditional QTEs and ATE can be identified from the result in Corollary 3.4. Specifically, one can derive the bounds on the unconditional QTE, as shown in Theorem 3.5 by inverting the bounds on the unconditional cdf. Once we identify the unconditional QTE for each quantile level $\tau \in (0, 1)$, the unconditional ATE is identified by taking average of the bounds over $\tau \in [0, 1]$.

3.3 Tightening Bounds: Stochastically Monotone Running Variable

It is well recognized that, in RD designs, the conditional expectation functions of potential outcomes on the running variable often exhibit monotonicity. Moreover, a growing body of empirical evidence provides support for the plausibility of such monotonicity in many applied settings (e.g., Lalive (2007); Lee (2008); Duflo et al. (2011); Abdulkadiroglu et al. (2014); Kaniel and Parham (2017)).⁸ In addition, it is well established in the literature that monotonicity assumptions can have substantial identifying power (cf. Manski (1997); Manski and Pepper (2000)).

We propose an assumption so called the *stochastically monotone running variable (SMRV)* to tighten the bounds on the counterfactual distribution functions. The SMRV assumption is motivated by the stochastically monotone instrumental variable assumption in Mourifie et al. (2020) and it places a distributional restriction among the conditional distributions of the potential outcomes on the running variable.

⁸Motivated by such empirical evidence, several studies have proposed estimation and inference procedures that explicitly incorporate monotonicity restrictions on the functions of interest in RD designs (e.g., Kwon and Kwon (2020); Babii and Kumar (2023)).

Assumption 3.3 (Stochastically Monotone Running Variable (SMRV)). *For any $r, \tilde{r} \in \text{Supp}(R)$ such that $r \geq \tilde{r}$, we have, for all $y \in \mathbb{R}$,*

$$F_{Y_d|R}(y|r) \leq F_{Y_d|R}(y|\tilde{r})$$

for $d \in \{0, 1\}$.

Assumption 3.3 imposes a first-order stochastic dominance restriction on the conditional distributions of the potential outcomes conditional on the running variable, with the distributions ordered according to the value of the running variable. Stochastic dominance assumptions are widely imposed in the literature and have proven useful for tightening identified bounds (e.g., [Blundell et al. \(2007\)](#); [Okumura and Usui \(2014\)](#); [Mourifie et al. \(2020\)](#); [Lee \(2024\)](#)). Note that the direction of monotonicity may be reversed depending on the empirical context. Moreover, such assumptions are subject to empirical testing.⁹ We note that under Assumptions 3.1 and 3.3, bounds on the conditional distribution functions can be derived using arguments analogous to those in Theorem 3.3.

Theorem 3.7. *Suppose that Assumptions 3.1 and 3.3 hold. For each $d \in \{0, 1\}$, let $r_d \in \mathcal{R}_d$ be given. Then, we have*

$$\begin{aligned} F_{Y_1|R}(y|r_0) &\in [LB_1^{SMRV}(k_{1,c}, r_0), UB_1^{SMRV}(k_{1,c}, r_0)], \\ F_{Y_0|R}(y|r_1) &\in [LB_0^{SMRV}(k_{1,c}, r_1), UB_0^{SMRV}(k_{1,c}, r_1)], \end{aligned}$$

where

$$\begin{aligned} LB_1^{SMRV}(k_{1,c}, r_0) &\equiv F_{Y_1|R}(y|r_1^*), \\ UB_1^{SMRV}(k_{1,c}, r_0) &\equiv \min \{F_{Y_1|R}(y|r_1^*) + k_{1,c} \cdot |r_0 - r_1^*|, 1\}, \\ LB_0^{SMRV}(k_{0,c}, r_1) &\equiv \max \{F_{Y_0|R}(y|r_0^*) - k_{0,c} \cdot |r_1 - r_0^*|, 0\}, \\ UB_0^{SMRV}(k_{0,c}, r_1) &\equiv F_{Y_0|R}(y|r_0^*), \end{aligned}$$

for all $y \in \mathbb{R}$.

It is straightforward to see that the bounds under Assumptions 3.1 and 3.3 are tighter than those in Theorem 3.3 in the sense that $UB_1^{SMRV}(k_{1,c}, r_0) \leq \overline{F}_{Y_1|R}^{k_{1,c}}(y | r_0)$ and $LB_0^{SMRV}(k_{0,c}, r_1) \geq$

⁹[Shen and Zhang \(2016\)](#) and [Qu and Yoon \(2019\)](#) propose inference methods for distributional treatment effects and, as part of their analysis, consider tests for stochastic dominance in RD designs.

$\underline{F}_{Y_0|R}^{k_0,c}(y \mid r_1)$, implying that the SMRV assumption has substantial identifying power.

Although we focus on identifying the distribution functions of the potential outcomes, our approach can also be applied to identifying their conditional mean functions, in a similar spirit to [Kim et al. \(2018\)](#). A mean version of Assumption 3.3 can therefore be proposed. We provide these results in Appendix C.

4 Estimation of Bounds

We consider estimation of the bounds on the (conditional) distribution functions of the potential outcomes, as well as the bounds on causal parameters such as the (conditional) ATE and QTE. Let $\{(Y_i, D_i, R_i) : i = 1, 2, \dots, n\}$ be an i.i.d. sample from the population (Y, D, R) . For each $d \in \{0, 1\}$, define $\mathcal{J}_d \equiv \{i : R_i \in \mathcal{R}_d\}$ and denote the cardinality of $\mathcal{J}_d$ by n_d . In addition, let $\mathcal{Y}_d$ denote the support of Y_d . Since the (conditional) ATE and QTE are functionals of the (conditional) distribution functions of the potential outcomes, we first focus on estimation of these distribution functions.

To this end, we employ the distributional regression (DR) considered by [Foresi and Peracchi \(1995\)](#) and [Chernozhukov et al. \(2013\)](#).¹⁰ Let $d \in \{0, 1\}$ be given. We then assume that for each $y \in \mathcal{Y}_d$ and $r \in \mathcal{R}_d$,

$$F_{Y_d|R}(y|r) = \Lambda \left(\mathcal{P}(r)' \beta_d(y) \right), \quad (4)$$

where $\mathcal{P}(r)$ is a vector of transformation of r such as polynomials or B-splines, $\Lambda(\cdot)$ is a known link function such as the standard logistic or normal distribution function. Note that the semiparametric specification of the conditional distribution in (4) may not hold for $r \in \mathcal{R}_{d'}$ with $d \neq d'$.

Let $\hat{\beta}_d(y)$ denote an estimator of β_d . It can be obtained by solving the following optimization problem:

$$\hat{\beta}_d(y) = \arg \max \sum_{i \in \mathcal{J}_d} \left[\mathbf{1}(Y_i \leq y) \log \left(\Lambda \left(\mathcal{P}(R_i)' b \right) \right) + \mathbf{1}(Y_i > y) \log \left(1 - \Lambda \left(\mathcal{P}(R_i)' b \right) \right) \right]$$

¹⁰Estimation can also be implemented using local nonparametric methods, such as local polynomial estimation. When $r_1^* = r_0^* = c$, then the novel estimation approach proposed by [Calonico et al. \(2014\)](#) can be used to estimate the bounds on the conditional expectation functions that are provided in Appendix (see section C). To estimate the bounds on the conditional distribution functions in Theorem 3.3, we can employ the approaches proposed by [Frandsen et al. \(2012\)](#) or [Shen and Zhang \(2016\)](#). Nevertheless, we use semiparametric DR estimators for several reasons. First, the convergence rate of the DR estimators is $\sqrt{n}$, whereas the convergence rate of nonparametric estimators is slower than $\sqrt{n}$. Second, if we are interested in estimating the unconditional ATE and QTE, we need to average the estimated distribution functions over a subset of $\text{Supp}(R)$. The semiparametric estimator can substantially reduce the computational burden relative to fully nonparametric methods and facilitate establishment of the asymptotic theory for the estimators of the unconditional ATE and QTE.

for each $y \in \mathcal{Y}_d$. Then, the DR estimator of $F_{Y_d|R}$ is

$$\hat{F}_{Y_d|R}(y|r) \equiv \Lambda \left(\mathcal{P}(r)' \hat{\beta}_d(y) \right).$$

Once we obtain an estimator of $F_{Y_d|R}(\cdot)$, it is straightforward to construct estimators of the (conditional) ATE and QTE. When the conditional distributions of the potential outcomes are identified as in (2), the corresponding τ -th quantile is partially identified as in (3). These bounds can also be obtained by inverting the lower and upper bounds on the conditional distribution functions. The bounds on the conditional ATE can be estimated by averaging the bounds on the CQTE. To estimate the unconditional ATE and QTE, we use the estimated conditional distributions. Once we estimate the conditional distributions, the unconditional distributions can be estimated by averaging over R :

$$\hat{F}_{Y_d}(y) \equiv \frac{1}{n_d} \sum_{i \in \mathcal{J}_d} \Lambda \left(\mathcal{P}(R_i)' \hat{\beta}_d(y) \right).$$

One can then use the estimated unconditional distributions to estimate the bounds on the ATE and QTE.

It is worth noting that, by virtue of Theorem 3.2, we do not need to account for the sample variation in $k_{d,c} |r_{d'} - r_d^*|$ for each $d \in \{0, 1\}$ and $d' = 1 - d$ as the points r_1^* and r_0^* are usually known in practice and $k_{d,c}$ is chosen by researchers for each $d \in \{0, 1\}$. Consequently, it is sufficient to consider the distributional approximation of the estimators of the (conditional) distribution functions of the potential outcomes.

5 Simulation

We conduct a Monte Carlo simulation to investigate the finite sample performance of the identification results. In the Monte Carlo simulation, the potential outcomes and treatment variable are generated as follows:

$$Y_1 = g_1(R) + s_1(R) \cdot \epsilon_1,$$

$$Y_0 = g_0(R) + s_0(R) \cdot \epsilon_0,$$

$$D = \mathbf{1}(R \geq 0),$$

where $(\epsilon_1, \epsilon_0)'|R \sim BVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.2 \\ 0.2 & 1 \end{pmatrix}\right)$, $R \sim Uniform(-1, 1)$, $g_1(r) = 2 + 0.5 \cdot r$, $g_0(r) = 1 + 0.3r$, $s_1(r) = 0.8 + (0.4 \cdot r)^2$ and $s_0(r) = 0.6 + (0.1 + 0.3 \cdot r)^2$. This data generating process (DGP) yields a location-scale shift quantile regression model, which generates different shapes of the conditional distribution (cf. [Koenker and Xiao \(2002\)](#)). We generate 5,000 observations for each simulation, and the number of replications is set to be 1,000.

To estimate the conditional distributions, we employ the DR with $\mathcal{P}(x) = x$. The grid size over $\mathcal{Y}_d$ is set to be 0.01, and we set $\mathcal{Y}_1 = \mathcal{Y}_d$ for simplicity. To construct the bounds on the conditional distributions, it is necessary to choose the nuisance parameters $k_{d,c}$ for $d \in \{0, 1\}$. Since the derivative of the conditional distribution over the region where it is identified is given by $\lambda(r \cdot \beta_d(y)) \beta_d(y)$, we set $k_{d,c}$ equal to the maximum absolute value of this derivative over the identified region.

Figure 5.1 reports the simulation results on the bounds on the conditional distributions of the potential outcomes. We consider four different conditioning values, $r \in \{\pm 0.25, \pm 0.5\}$. The blue lines are pointwise average of the upper bounds on $F_{Y_1|R}(\cdot|r)$ for $r \in \{-0.5, -0.25\}$ over the simulation repetitions, and the red lines depict pointwise average of the lower bounds on $F_{Y_0|R}(\cdot|r)$ for $r \in \{0.25, 0.5\}$. We can find that the bounds contain the true distribution function in all cases. In addition, the (average) upper bounds on $F_{Y_1|R}$ are very close to the true conditional distribution over some region of $\mathcal{Y}_1$. A similar pattern can be found for the lower bounds on $F_{Y_0|R}$.

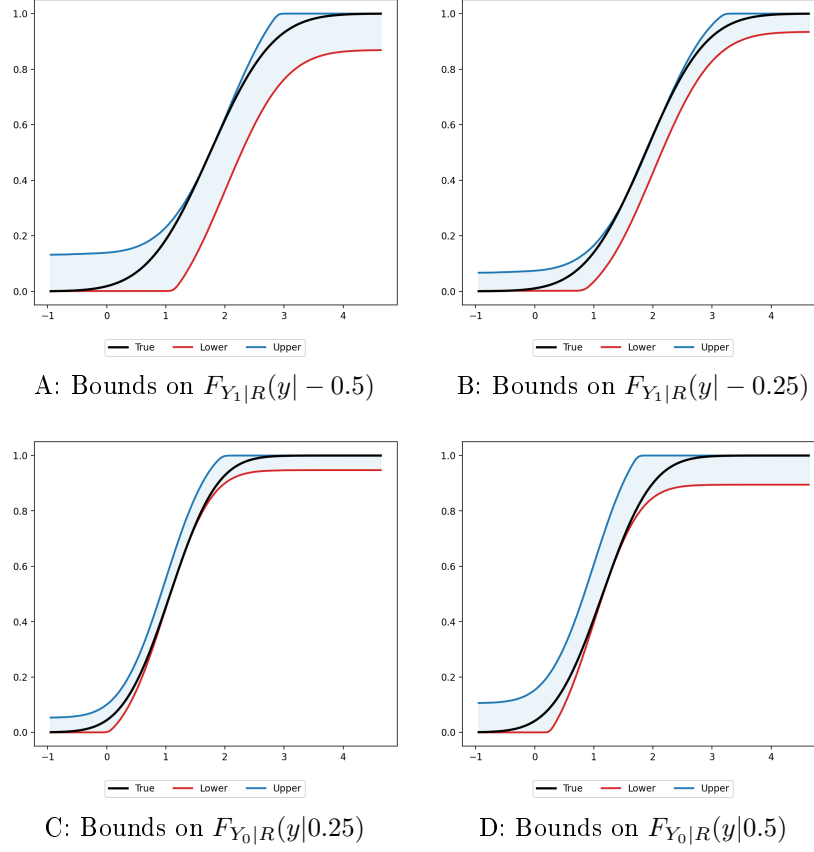
Figure 5.2 show the estimated bounds on the unconditional distribution functions F_{Y_1} and F_{Y_0} . As explained earlier, these bounds are obtained by taking average of the conditional distribution functions over $Supp(R)$. We can see that the bounds on the unconditional distributions are much tighter than those on the conditional distributions, and this is consistent with the argument in [Firpo and Ridder \(2019\)](#).¹¹

We report the simulation results for causal parameters such as the CQTE, CATE, QTE, and ATE in Table 5.1. All estimated bounds contain the true value of the corresponding causal parameter. We also observe that the upper bounds tend to be farther from the true parameter values than the lower bounds.

We investigate the identifying power of the SMRV assumption. The DGP is modified slightly so that it satisfies the SMRV assumption. Specifically, we set $s_1(r) = s_0(r) = 1$, which yields a location shift model that satisfies the SMRV assumption. Tables 5.2 and 5.3 report the simulation results for

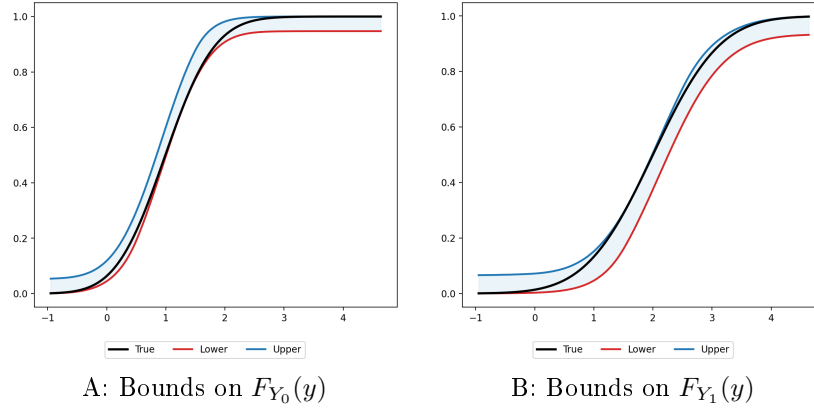
¹¹[Firpo and Ridder \(2019\)](#) show that when the potential outcomes are correlated with an observed variable, the bounds on the (unconditional) distribution of treatment effects can be tightened by averaging over the values of that covariate. Although the role of the running variable in sharp RD designs differs from that of observed covariates in randomized experiments, the tightening of the bounds on the unconditional distributions of the potential outcomes in our framework may reflect a similar idea.

Figure 5.1: Bounds on Conditional CDFs: Location-Scale Shift Model



Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

Figure 5.2: Bounds on Unconditional CDFs: Location-Scale Shift Model



Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

Table 5.1: Bounds on Causal Parameters: Location-Scale Shift Model

Type	τ	r	True	Estimated bounds
<i>CQTE</i>				
CQTE	0.1	-0.5	0.596	[-1.002, 1.358]
CQTE	0.1	0.5	0.873	[0.648, 2.080]
CQTE	0.3	-0.5	0.775	[0.723, 1.342]
CQTE	0.3	0.5	1.007	[0.943, 1.314]
CQTE	0.5	-0.5	0.900	[0.896, 1.425]
CQTE	0.5	0.5	1.100	[1.084, 1.405]
CQTE	0.7	-0.5	1.025	[0.980, 1.597]
CQTE	0.7	0.5	1.193	[1.171, 1.544]
CQTE	0.9	-0.5	1.204	[0.963, 3.155]
CQTE	0.9	0.5	1.327	[-1.458, 1.836]
<i>QTE</i>				
QTE	0.1	-	0.727	[0.353, 1.396]
QTE	0.3	-	0.888	[0.829, 1.328]
QTE	0.5	-	1.000	[0.979, 1.416]
QTE	0.7	-	1.112	[1.068, 1.573]
QTE	0.9	-	1.274	[1.091, 2.124]
<i>CATE</i>				
CATE	-	-0.5	0.900	[0.661, 1.644]
CATE	-	0.5	1.100	[0.844, 1.521]
<i>ATE</i>				
ATE	-	-	1.000	[0.844, 1.557]

Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

Table 5.2: Identifying Power of the SMRV Assumption: CQTE

τ	CQTE at $r = -0.5$			CQTE at $r = 0.5$		
	True	Baseline	SMRV	True	Baseline	SMRV
0.1	0.900	[-0.753, 1.564]	[-0.753, 1.109]	1.100	[0.966, 1.717]	[0.966, 1.252]
0.3	0.900	[0.813, 1.406]	[0.813, 1.135]	1.100	[1.083, 1.432]	[1.083, 1.249]
0.5	0.900	[0.885, 1.396]	[0.885, 1.142]	1.100	[1.096, 1.398]	[1.096, 1.248]
0.7	0.900	[0.875, 1.465]	[0.875, 1.147]	1.100	[1.062, 1.411]	[1.062, 1.244]
0.9	0.900	[0.711, 2.977]	[0.711, 1.154]	1.100	[0.777, 1.529]	[0.777, 1.244]

Note: The nuisance parameters are set to $k_0 = 0.120$ and $k_1 = 0.199$. The true DGP is a location shift model, where $s_1(r) = s_0(r) = 1$.

the CQTE and QTE. The baseline bounds are estimated without imposing the SMRV assumption, whereas the SMRV bounds are estimated under the SMRV assumption. It is evident that the bounds under the SMRV assumption contain the corresponding causal parameter and are much tighter than the baseline bounds. For all causal parameters, the upper bounds are substantially lower, resulting in much narrower bounds. The lower bounds under the SMRV assumption are the same to the baseline lower bounds, which reflects the fact that the SMRV assumption imposes monotonicity on the conditional distributions only in one direction. Nevertheless, the simulation results confirm that the SMRV assumption has substantial identifying power.

Table 5.3: Identifying Power of the SMRV Assumption: QTE

τ	True	Baseline	SMRV
0.1	1.000	[0.636, 1.613]	[0.636, 1.169]
0.3	1.000	[0.948, 1.414]	[0.948, 1.185]
0.5	1.000	[0.991, 1.398]	[0.991, 1.193]
0.7	1.000	[0.973, 1.440]	[0.973, 1.203]
0.9	1.000	[0.812, 1.740]	[0.812, 1.214]

Note: The nuisance parameters are set to $k_0 = 0.120$ and $k_1 = 0.199$. The true DGP is a location shift model, where $s_1(r) = s_0(r) = 1$.

6 Empirical Application

We provide an empirical illustration that considers the effect of better schools on student's academic performance to demonstrate how the proposed methods can be applied in practice. When considering the effect of attending a better school on student performance, it is well documented in the literature that the effect for students who are marginally admitted to the school may differ from the effect for those who qualify more easily (e.g., [Abdulkadiroğlu et al. \(2014\)](#); [Dobbie and](#)

Fryer Jr (2014); Angrist and Rokkanen (2015)). For this reason, it is worthwhile to investigate the causal effects at values of the running variable away from the cutoff, and we apply the proposed methods to shed light on potential heterogeneity in the causal effect across the running variable.

We revisit the empirical study of Pop-Eleches and Urquiola (2013), who examine the effect of attending a better school on student performance in the Romanian secondary school system. In Romania, admission from middle school to high school is determined by students’ transition scores, which are based on their performance on a national eighth-grade exam, together with their ranked lists of preferred high schools. As described in Pop-Eleches and Urquiola (2013), this assignment process generates a cutoff score for each school, which makes it possible to use an RD design to identify the causal effect of access to higher-achieving schools. Among the outcomes they consider is students’ score on the Baccalaureate exam. This outcome is particularly important because students must pass the exam to be eligible to apply to university, and the exam score is also used as a key factor in the admissions process.

We use the dataset from Shen and Zhang (2016), which is identical to that used in Pop-Eleches and Urquiola (2013), except that Shen and Zhang (2016) restrict the sample to two-school towns to mitigate estimation bias caused by multiple cutoff values. The outcome variable Y is the grade of the Baccalaureate exam, and the running variable R is the transition score centered at the cutoff value. Table 6.1 reports the summary statistics of the dataset, and Figure A.1 depict the kernel density estimators for R and Y . While the cutoff value is set to be zero, more than half of students in the sample (approximately 72%) are above the cutoff value. The distribution of the transition score is not symmetric around the cutoff value, and it is not uniformly distributed.

Table 6.1: Summary Statistics

Variable	n	Mean	Std.dev	Min	Q25	Q50	Q75	Max
Baccalaureate grade (Y)	14,765	-0.06	0.80	-2.95	-0.62	0.01	0.55	2.00
Transition score (R)	14,765	0.48	0.75	-1.93	-0.09	0.49	1.07	1.93
Treatment (D)	14,765	0.72	0.45	0.00	0.00	1.00	1.00	1.00

Note: The running variable is the transition score centered at the admission cutoff, so that the cutoff value is zero; that is, $D = \mathbf{1}(R \geq 0)$. The Baccalaureate exam grade is the outcome variable.

The main ingredients to estimate the bounds are the conditional distributions of the potential outcomes on the running variable, and the distribution functions are evaluated at the boundary points of $\mathcal{R}_1$ and $\mathcal{R}_0$. As in the simulation study, we consider $\mathcal{R}_1 = [c, r_{max}]$ and $\mathcal{R}_0 = [r_{min}, c]$, and thus, the boundary points are the cutoff level c . We normalize the cutoff value to zero. Since our extrapolation is based on the boundary points of $\mathcal{R}_1$ and $\mathcal{R}_0$, which are both equal to c , we first

compare the DR estimators evaluated at $R = c$ with the corresponding nonparametric estimators proposed by Calonico et al. (2014). As shown in Figure A.2 in Appendix, the difference between these two estimators is negligible. This comparison shows that the empirical results are not sensitive to the choice of estimation methods.

We focus on the identification and estimation of causal parameters, including the CQTE, QTE, CATE, and ATE. Table 6.2 reports the estimated bounds on the CQTE for different quantile levels and values of the running variable. To facilitate comparison, we also report the RD estimates of the QTE at the cutoff, that is, the CQTE at $R = 0$. In addition, we compare the estimated bounds obtained with and without the SMRV assumption. For $R = r$ such that $r < 0$, the estimated bounds contain the RD estimates in all cases. Moreover, the bounds include zero, so we cannot determine the sign of the CQTE conditional on $R = r$ when $r = 0$. In contrast, for $R = r$ such that $r > 0$, some estimated bounds on the CQTE conditional on $R = r$ do not contain the RD estimate of the QTE at the cutoff. For example, the estimated lower bound on the 0.1th CQTE given $R = 0.5$ is 0.227 both with and without the SMRV assumption, whereas the RD estimate at the cutoff is 0.182. Similar results can be found for different quantile levels. The estimated lower bounds on the τ -th CQTE given $R = 0.5$ are greater than the RD estimate at the cutoff except for $\tau \in \{0.8, 0.9\}$. Setting aside sampling error, this result suggests that the 0.1th CQTE given $R = 0.5$ differs from the QTE at the cutoff for the same quantile level. Similar patterns are observed for other values of r . In particular, the lower bounds tend to increase as the conditioning value r increases.

Figure 6.1 plots the estimated bounds, with and without the SMRV assumption, for each fixed quantile level. The figure clearly shows that the SMRV assumption does not improve the lower bounds, but it substantially improves the upper bounds, thereby yielding narrower identified sets. Consistent with the results in Table 6.2, these findings suggest that the effect of attending a better school on performance on the Baccalaureate exam may be heterogeneous across subpopulations defined by the value of the running variable, as some bounds on the CQTE at values of the running variable different from the cutoff do not contain the corresponding RD estimates.

In Figure A.3 in Appendix, we report the estimated bounds on the CQTE for various values of k_0 and k_1 to see the sensitivity of the bounds to these nuisance parameters. Consistent with our theoretical predictions, the bounds become wider as either k_0 or k_1 increases.

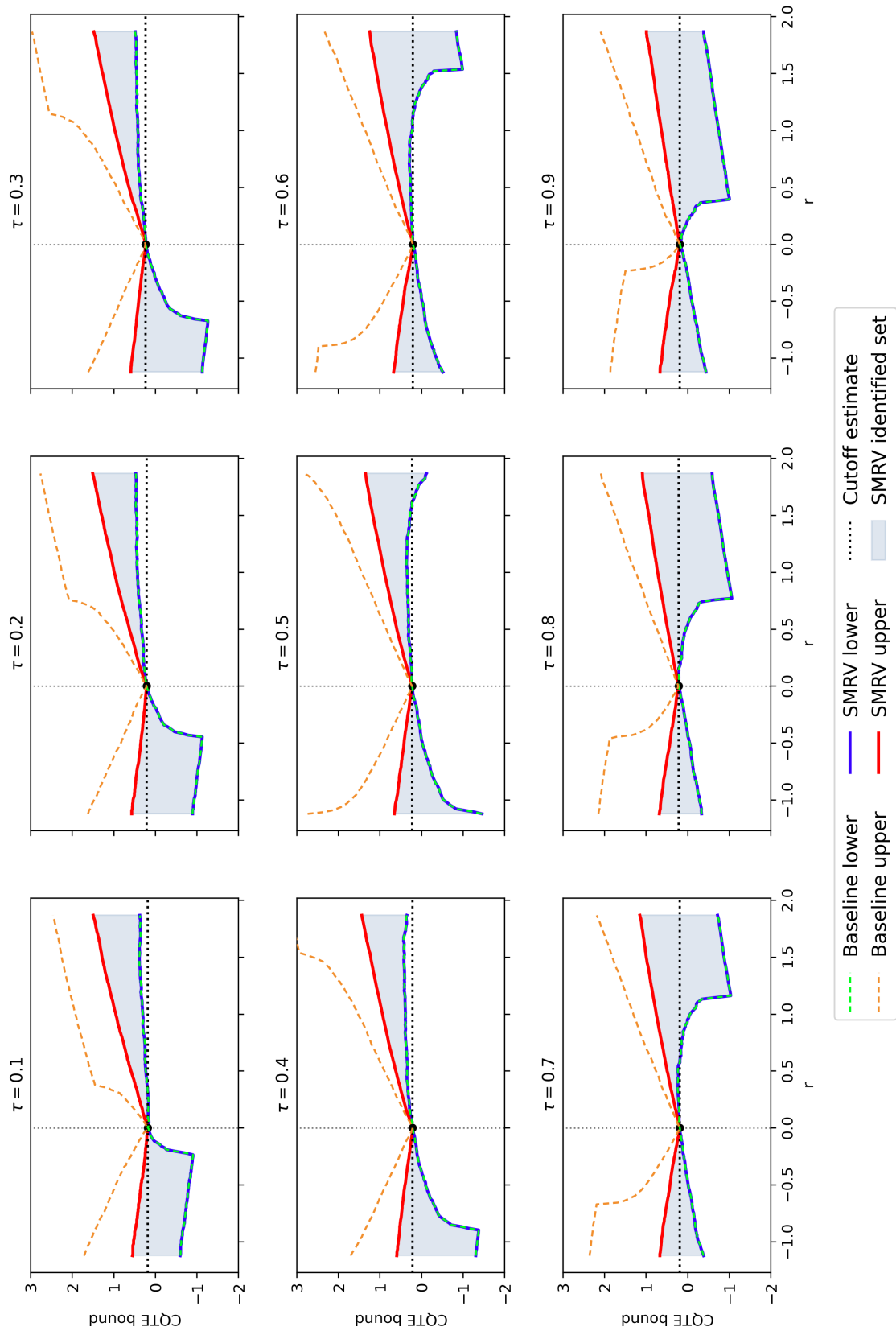
We now consider the estimated bounds on the unconditional QTE. Table 6.3 presents the estimated bounds on the unconditional QTE as well as the RD estimate of the QTE at $R = 0$. While the bounds at $\tau \in \{0.1, 0.9\}$ are uninformative in the sense that the sign of QTE is not identified, we can identify the sign of QTE at other quantile levels. An interesting finding is that, for τ ranging

Table 6.2: Bounds on CQTE

τ	CQTE at $r = 0$	CQTE at $r = -1$		CQTE at $r = -0.5$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	[-0.627, 1.575]	[-0.627, 0.522]	[-0.807, 0.953]	[-0.807, 0.342]
0.2	0.208	[-0.930, 1.474]	[-0.930, 0.534]	[-1.112, 0.833]	[-1.112, 0.352]
0.3	0.232	[-1.159, 1.451]	[-1.159, 0.559]	[-0.247, 0.838]	[-0.247, 0.401]
0.4	0.211	[-1.343, 1.510]	[-1.343, 0.557]	[-0.106, 0.849]	[-0.106, 0.393]
0.5	0.220	[-0.538, 1.710]	[-0.538, 0.602]	[-0.025, 0.863]	[-0.025, 0.417]
0.6	0.201	[-0.365, 2.499]	[-0.365, 0.619]	[-0.015, 0.911]	[-0.015, 0.439]
0.7	0.183	[-0.269, 2.315]	[-0.269, 0.628]	[-0.012, 1.005]	[-0.012, 0.423]
0.8	0.212	[-0.288, 2.103]	[-0.288, 0.639]	[-0.053, 1.888]	[-0.053, 0.425]
0.9	0.182	[-0.372, 1.810]	[-0.372, 0.616]	[-0.107, 1.629]	[-0.107, 0.435]
τ	CQTE at $r = 0$	CQTE at $r = 0.5$		CQTE at $r = 1$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	[0.227, 1.551]	[0.227, 0.582]	[0.299, 1.917]	[0.299, 0.948]
0.2	0.208	[0.332, 1.028]	[0.332, 0.620]	[0.427, 2.241]	[0.427, 0.984]
0.3	0.232	[0.364, 0.924]	[0.364, 0.632]	[0.424, 1.784]	[0.424, 0.972]
0.4	0.211	[0.342, 0.882]	[0.342, 0.614]	[0.382, 1.521]	[0.382, 0.934]
0.5	0.220	[0.306, 0.841]	[0.306, 0.589]	[0.339, 1.429]	[0.339, 0.890]
0.6	0.201	[0.268, 0.814]	[0.268, 0.532]	[0.226, 1.347]	[0.226, 0.811]
0.7	0.183	[0.232, 0.753]	[0.232, 0.489]	[-0.058, 1.301]	[-0.058, 0.750]
0.8	0.212	[0.039, 0.731]	[0.039, 0.477]	[-0.961, 1.241]	[-0.961, 0.716]
0.9	0.182	[-0.959, 0.782]	[-0.959, 0.419]	[-0.751, 1.246]	[-0.751, 0.627]
τ	CQTE at $r = 0$	CQTE at $r = 1.25$		CQTE at $r = 1.5$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	[0.337, 2.084]	[0.337, 1.115]	[0.381, 2.252]	[0.381, 1.282]
0.2	0.208	[0.442, 2.394]	[0.442, 1.137]	[0.448, 2.544]	[0.448, 1.286]
0.3	0.232	[0.439, 2.604]	[0.439, 1.118]	[0.451, 2.749]	[0.451, 1.264]
0.4	0.211	[0.406, 1.892]	[0.406, 1.085]	[0.413, 1.584]	[0.413, 1.218]
0.5	0.220	[0.340, 1.707]	[0.340, 1.022]	[0.271, 2.042]	[0.271, 1.163]
0.6	0.201	[0.135, 1.623]	[0.135, 0.949]	[-0.219, 1.906]	[-0.219, 1.074]
0.7	0.183	[-0.998, 1.547]	[-0.998, 0.871]	[-0.879, 1.801]	[-0.879, 0.990]
0.8	0.212	[-0.851, 1.507]	[-0.851, 0.826]	[-0.751, 1.739]	[-0.751, 0.926]
0.9	0.182	[-0.647, 1.476]	[-0.647, 0.731]	[-0.553, 1.736]	[-0.553, 0.825]

Note: The cutoff value is zero. “Baseline” and “SMRV” denote the corresponding bounds on the CQTE given $R = r$ without and with the SMRV assumption, respectively. The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

Figure 6.1: Bounds on the CQTE for Different Values of $R = r$



from 0.2 to 0.7, the lower bounds are greater than the RD estimates. This implies that the QTE of attending a better school on Bacculaureate grades for *the whole population* may differ from that for *students at the margin* of admission to a better high school.

Table 6.3: Bounds on the QTE

τ	RD Estimate	QTE	
		Baseline	SMRV
0.1	0.182	$[-0.117, 1.668]$	$[-0.117, 0.524]$
0.2	0.208	$[0.119, 1.331]$	$[0.119, 0.601]$
0.3	0.232	$[0.195, 1.159]$	$[0.195, 0.660]$
0.4	0.211	$[0.244, 1.142]$	$[0.244, 0.683]$
0.5	0.220	$[0.257, 1.113]$	$[0.257, 0.698]$
0.6	0.201	$[0.254, 1.110]$	$[0.254, 0.666]$
0.7	0.183	$[0.241, 1.114]$	$[0.241, 0.644]$
0.8	0.212	$[0.014, 1.108]$	$[0.014, 0.654]$
0.9	0.182	$[-0.846, 1.186]$	$[-0.846, 0.596]$

Note: The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

Using the estimated bounds on the CQTE and QTE, we derive bounds on the CATE and ATE. The RD estimate, which corresponds to the ATE given $R = 0$, is 0.2. The estimated bounds exhibit heterogeneity across values of the running variable. In particular, the sign of the CATE is identified for $r \in \{0.5, 1\}$, and the sign of the ATE for the whole population is also identified. Nevertheless, we do not find evidence that the ATE for students at the margin of admission to a better high school differs from that for other subpopulation of students.

Table 6.4: Bounds on the CATE and ATE

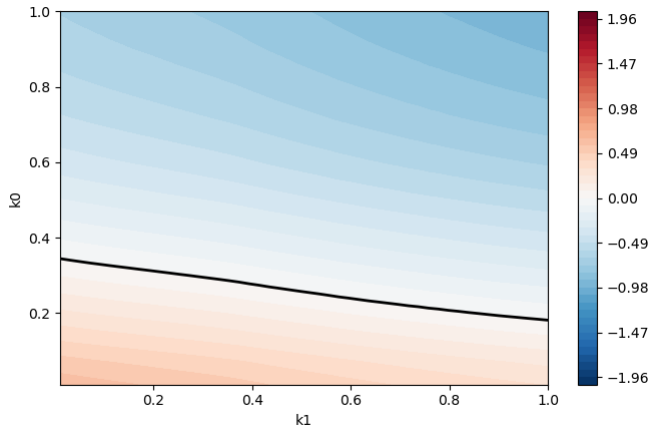
	r	Baseline	SMRV
CATE	-1	$[-0.668, 1.833]$	$[-0.668, 0.587]$
CATE	-0.5	$[-0.270, 1.074]$	$[-0.270, 0.403]$
CATE	0.5	$[0.132, 0.919]$	$[0.132, 0.544]$
CATE	1	$[0.023, 1.568]$	$[0.023, 0.843]$
CATE	1.25	$[-0.028, 1.864]$	$[-0.028, 0.976]$
CATE	1.5	$[-0.076, 2.147]$	$[-0.076, 1.110]$
ATE	-	$[0.010, 1.230]$	$[0.010, 0.640]$

Note: The RD estimate is 0.2. The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

We then conduct a breakdown frontier analysis for the ATE. The purpose of breakdown frontier analysis is to determine the maximal degree to which the underlying assumptions can be relaxed while still preserving a given conclusion.¹² We seek the combinations of k_0 and k_1 that yield the

¹²One can refer to [Masten and Poirier \(2020\)](#), who develop an approach to inference for breakdown frontiers for relaxation of the random assignment of treatment assumption and rank invariance, for a detailed discussion on breakdown frontier analysis.

Figure 6.2: Breakdown Frontier for $ATE \geq 0$



Note: The x -axis indexes the values of k_1 , and y -axis indexes the values of k_0 . The red area is the set of (k_0, k_1) such that the estimated lower bound on ATE is greater than or equal to zero. The blue area is the set of (k_0, k_1) such that the estimated lower bound on ATE is less than zero. The black solid line is the breakdown frontier.

lower bound on the ATE for the whole population is greater than or equal to zero. In this case, the breakdown frontier can be characterized by the set of values of (k_0, k_1) such that the lower bound on the ATE is greater than or equal to zero.

Figure 6.2 presents the breakdown frontier for the conclusion $ATE \geq 0$. The red region represents the set of (k_0, k_1) values for which the estimated lower bound on the ATE is nonnegative. The blue region represents the set of (k_0, k_1) values for which the estimated lower bound on the ATE is negative. Accordingly, the black solid line in the figure denotes the breakdown frontier.

To maintain the conclusion that $ATE \geq 0$, the maximum value of k_0 when the potential outcome Y_1 and the running variable R are independent (i.e., $k_1 = 0$), is approximately 0.35. On the other hand, when we do not impose any restriction on the dependence between Y_1 and R (i.e., $k_1 = 1$), the maximal relaxation of the independence between Y_0 and R is about 0.2. The breakdown frontier in Figure 6.2 clearly illustrates the trade-off between the two assumptions on the dependence between the potential outcomes and the running variable required to maintain a given conclusion.

7 Conclusions

In this paper, we propose a framework for extrapolating RD treatment effects to a broader class of causal parameters in sharp RD designs. We adopt a partial identification strategy that delivers bounds on the causal parameters of interest. The key identifying assumption is a smoothness re-

striction, parameterized through sensitivity measures that capture the degree of smoothness of the conditional distribution functions of the potential outcomes on the running variable. Under this framework, we derive sharp bounds on the counterfactual means and distributions of the potential outcomes that are tractable and straightforward to estimate. The proposed approach is well suited to sensitivity analysis in sharp RD designs, and the Monte Carlo simulations provide support for the theoretical results developed in this paper. We further present an empirical application to illustrate the practical relevance of these theoretical findings. The results of the empirical application suggest that RD estimates need not be representative of the relevant causal parameters for broader subpopulations or even for the whole population.

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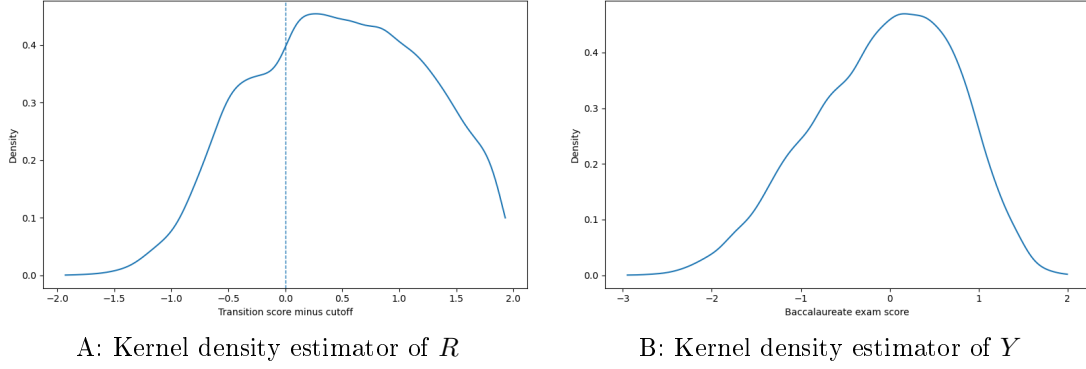
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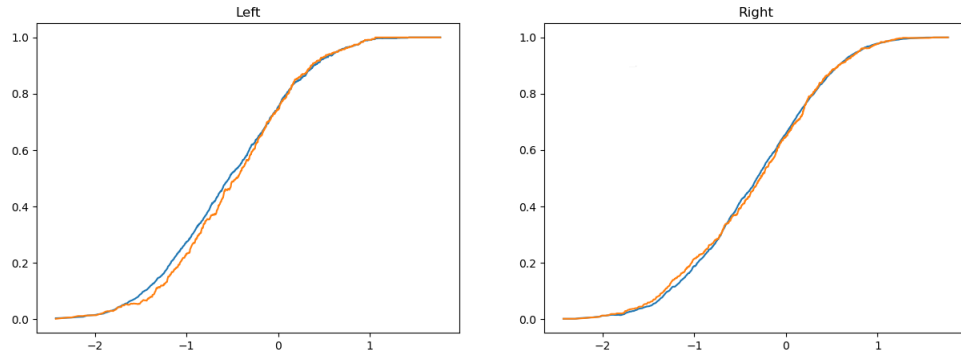
A Additional Results for the Empirical Application

Figure A.1: Kernel Density Estimators of the Outcome and Running Variables



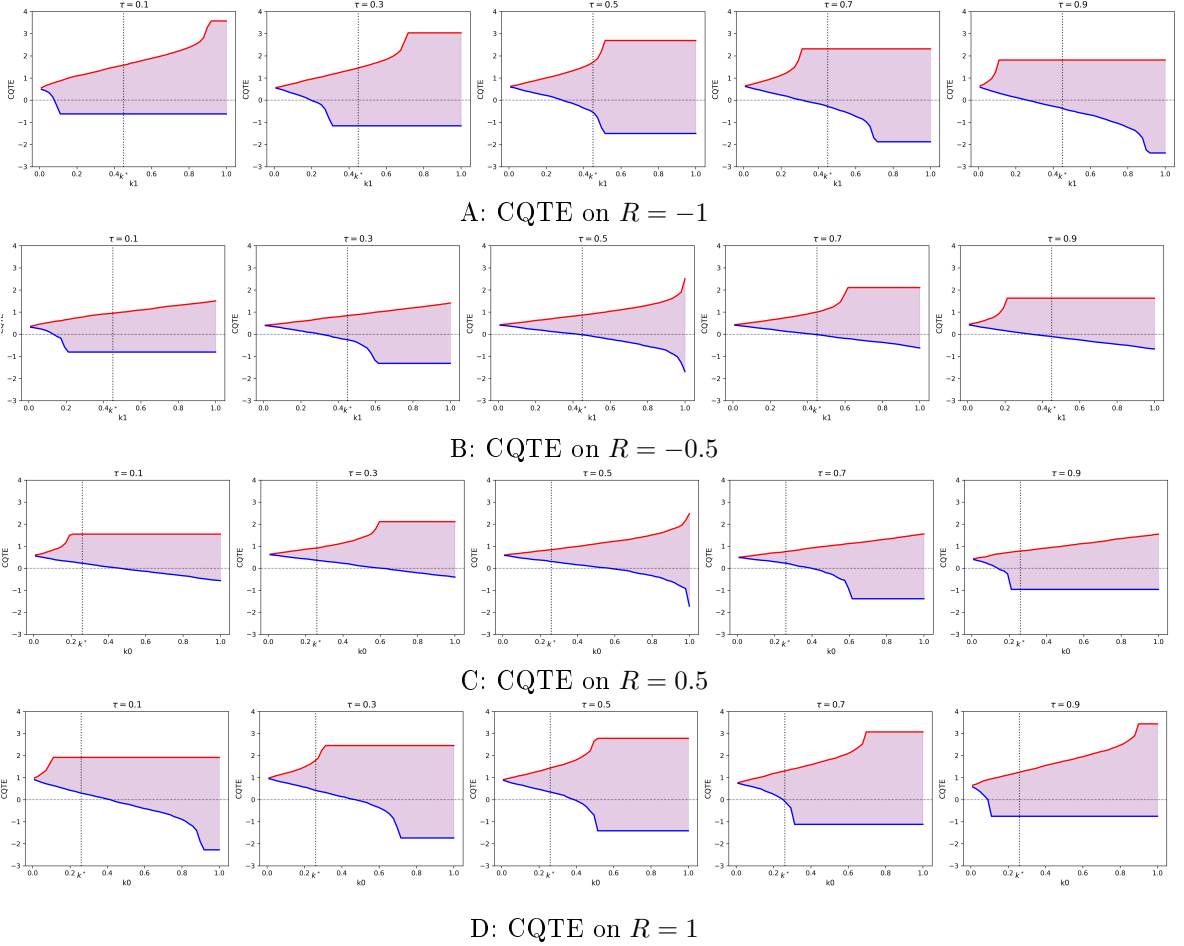
Note: We use the Gaussian kernel for both cases. The bandwidths are set to be 0.110 and 0.117 for R and Y , respectively.

Figure A.2: Comparison of DR and Nonparametric Estimators of $F_{Y_1|R}(\cdot|c)$ and $F_{Y_0|R}(\cdot|c)$



Note: The blue lines are the DR estimators, and the orange lines are the nonparametric local linear estimators of the distribution functions.

Figure A.3: Sensitivity of Bounds on the CQTE



Note: We consider $\tau \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ to estimate the bounds on CQTE. k^* denotes the value of k_0 or k_1 , estimated as the maximum derivative of the conditional distribution function over its identified region. Specifically, $k^* = 0.45$ for $R = r < 0$, and $k^* = 0.26$ for $R = r > 0$.

B Mathematical Proofs

B.1 Proof of Theorem 3.1

Proof. We only prove the results on the lower and upper bounds on $E[h(Y_1)|R=r]$. The bounds on $E[h(Y_0)|R=r]$ can be proven by a similar way.

Let $r, \tilde{r} \in \text{Supp}(R)$ be given. Note that under Assumption 2.2, we have

$$E[h(Y_1)|R=\tilde{r}] - k_{1,h}|r - \tilde{r}| \leq E[h(Y_1)|R=r] \leq E[h(Y_1)|R=\tilde{r}] + k_{1,h}|r - \tilde{r}|.$$

By Assumption 2.1, $E[h(Y_1)|R=\tilde{r}]$ is identified for all $\tilde{r} \in \mathcal{R}_1$. Taking the supremum over $\tilde{r} \in \mathcal{R}_1$ to the both sides of $E[h(Y_1)|R=\tilde{r}] - k_{1,h}|r - \tilde{r}| \leq E[h(Y_1)|R=r]$ yields the lower bound. Similarly, the upper bound can be obtained by applying the infimum over $\tilde{r} \in \mathcal{R}_1$ to $E[h(Y_1)|R=r] \leq E[h(Y_1)|R=\tilde{r}] + k_{1,h}|r - \tilde{r}|$. $\blacksquare$

B.2 Proof of Theorem 3.2

Lemma B.1. *Suppose that Assumptions 2.1 and 2.2 hold, and let $d \in \{0, 1\}$ be given. Let h be a measurable function such that $E[|h(Y_d)|] < \infty$. Define*

$$E[h(Y_d) | R = r_d^*] \equiv \lim_{t \rightarrow r_d^*: t \in \mathcal{R}_d} E[h(Y_d) | R = t].$$

Then, $E[h(Y_d) | R = r_d^]$ is well-defined.*

Proof. Let $m_{d,h}(r) \equiv E[h(Y_d) | R = r]$ for $r \in \text{Supp}(R)$. Assumption 2.1 implies that $m_{d,h}(r)$ is identified for $r \in \mathcal{R}_d$. By Assumption 2.2, for any $r, \tilde{r} \in \text{Supp}(R)$,

$$|m_{d,h}(r) - m_{d,h}(\tilde{r})| \leq k_{d,h}|r - \tilde{r}|.$$

We first prove existence. For any sequence $(t_n) \subset \mathcal{R}_d$ with $t_n \rightarrow r_d^*$,

$$|m_{d,h}(t_n) - m_{d,h}(t_m)| \leq k_{d,h}|t_n - t_m| \rightarrow 0$$

as $n, m \rightarrow \infty$, so $(m_{d,h}(t_n))$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, there exists $m^* \in \mathbb{R}$ such that

$$|m_{d,h}(t_n) - m^*| \rightarrow 0.$$

By the definition above, $E[h(Y_d) \mid R = r_d^*] = m^*$.

We now prove uniqueness. Let $(t_n) \subset \mathcal{R}_d$ and $(s_m) \subset \mathcal{R}_d$ be sequences such that $t_n \rightarrow r_d^*$ and $s_m \rightarrow r_d^*$. By the existence result, there exist $m^*, \tilde{m}^* \in \mathbb{R}$ such that $m_{d,h}(t_n) \rightarrow m^*$ and $m_{d,h}(s_m) \rightarrow \tilde{m}^*$. For all $n, m \in \mathbb{N}$,

$$|m^* - \tilde{m}^*| \leq |m^* - m_{d,h}(t_n)| + |m_{d,h}(t_n) - m_{d,h}(s_m)| + |m_{d,h}(s_m) - \tilde{m}^*|.$$

Taking $\limsup_{n,m \rightarrow \infty}$ yields

$$|m^* - \tilde{m}^*| \leq \limsup_{n,m \rightarrow \infty} |m_{d,h}(t_n) - m_{d,h}(s_m)|.$$

By Assumption 2.2, it follows that

$$|m_{d,h}(t_n) - m_{d,h}(s_m)| \leq k_{d,h}|t_n - s_m|$$

for all $n, m \in \mathbb{N}$. Thus,

$$|m^* - \tilde{m}^*| \leq \limsup_{n,m \rightarrow \infty} k_{d,h}|t_n - s_m| = 0,$$

since $t_n \rightarrow r_d^*$ and $s_m \rightarrow r_d^*$, implying $m^* = \tilde{m}^*$. ■

Proof of Theorem 3.2

Proof. We only prove $LB_1(k_{1,h}, r_0)$ and $UB_1(k_{1,h}, r_0)$, and the other results follow by the same argument. Fix $r_0 \in \mathcal{R}_0$. By Lemma B.1, $E[h(Y_1) \mid R = r_1^*]$ is well-defined.

Lower bound. Take any $\tilde{r} \in \mathcal{R}_1$. By Assumption 2.2,

$$E[h(Y_1) \mid R = \tilde{r}] \leq E[h(Y_1) \mid R = r_1^*] + k_{1,h}|\tilde{r} - r_1^*|.$$

Since $r_0 < r_1^* \leq \tilde{r}$, we have $|r_0 - \tilde{r}| = |r_0 - r_1^*| + |\tilde{r} - r_1^*|$. Therefore,

$$\begin{aligned} E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}|r_0 - \tilde{r}| &\leq E[h(Y_1) \mid R = r_1^*] + k_{1,h}|\tilde{r} - r_1^*| - k_{1,h}(|r_0 - r_1^*| + |\tilde{r} - r_1^*|) \\ &= E[h(Y_1) \mid R = r_1^*] - k_{1,h}|r_0 - r_1^*|, \end{aligned}$$

which implies that $E[h(Y_1) \mid R = r_1^*] - k_{1,h}|r_0 - r_1^*|$ is an upper bound for $E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}|r_0 -$

$\tilde{r}$ for all $\tilde{r} \in \mathcal{R}_1$. Taking the supremum over $\tilde{r} \in \mathcal{R}_1$ yields

$$\sup_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}|r_0 - \tilde{r}|\} \leq E[h(Y_1) \mid R = r_1^*] - k_{1,h}|r_0 - r_1^*|. \quad (5)$$

To show the reverse inequality, take any sequence $(r_{1n}) \subset \mathcal{R}_1$ such that $r_{1n} \rightarrow r_1^*$. Then, for all $n \in \mathbb{N}$,

$$\sup_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}|r_0 - \tilde{r}|\} \geq E[h(Y_1) \mid R = r_{1n}] - k_{1,h}|r_0 - r_{1n}|.$$

Letting $n \rightarrow \infty$ and using Lemma B.1 together with $|r_0 - r_{1n}| \rightarrow |r_0 - r_1^*|$, we obtain

$$\sup_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}|r_0 - \tilde{r}|\} \geq E[h(Y_1) \mid R = r_1^*] - k_{1,h}|r_0 - r_1^*|. \quad (6)$$

Combining (5) and (6) yields

$$LB_1(k_{1,h}, r_0) = E[h(Y_1) \mid R = r_1^*] - k_{1,h}|r_0 - r_1^*|.$$

Upper bound. Take any $\tilde{r} \in \mathcal{R}_1$. By Assumption 2.2,

$$E[h(Y_1) \mid R = \tilde{r}] \geq E[h(Y_1) \mid R = r_1^*] - k_{1,h}|\tilde{r} - r_1^*|.$$

Since $r_0 < r_1^* \leq \tilde{r}$, we have $|r_0 - \tilde{r}| = |r_0 - r_1^*| + |\tilde{r} - r_1^*|$. Thus,

$$\begin{aligned} E[h(Y_1) \mid R = \tilde{r}] + k_{1,h}|r_0 - \tilde{r}| &\geq E[h(Y_1) \mid R = r_1^*] - k_{1,h}|\tilde{r} - r_1^*| + k_{1,h}(|r_0 - r_1^*| + |\tilde{r} - r_1^*|) \\ &= E[h(Y_1) \mid R = r_1^*] + k_{1,h}|r_0 - r_1^*|. \end{aligned}$$

Taking the infimum over $\tilde{r} \in \mathcal{R}_1$ yields

$$\inf_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] + k_{1,h}|r_0 - \tilde{r}|\} \geq E[h(Y_1) \mid R = r_1^*] + k_{1,h}|r_0 - r_1^*|. \quad (7)$$

To show the reverse inequality, take any sequence $(r_{1n}) \subset \mathcal{R}_1$ with $r_{1n} \rightarrow r_1^*$. Then, for all $n \in \mathbb{N}$,

$$\inf_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] + k_{1,h}|r_0 - \tilde{r}|\} \leq E[h(Y_1) \mid R = r_{1n}] + k_{1,h}|r_0 - r_{1n}|.$$

Letting $n \rightarrow \infty$ and using Lemma B.1 together with $|r_0 - r_{1n}| \rightarrow |r_0 - r_1^*|$ yields

$$\inf_{\tilde{r} \in \mathcal{R}_1} \{E[h(Y_1) \mid R = \tilde{r}] + k_{1,h}|r_0 - \tilde{r}|\} \leq E[h(Y_1) \mid R = r_1^*] + k_{1,h}|r_0 - r_1^*|. \quad (8)$$

Combining (7) and (8) proves the claim. ■

B.3 Proof of Theorem 3.3

We provide two lemmas that will be used for the proof of Theorem 3.3.

Lemma B.2. *Suppose that Assumption 3.1 holds and let $d \in \{0, 1\}$ be given. Define*

$$F_{Y_d|R}(y|r_d^*) \equiv \lim_{t \rightarrow r_d^*: t \in \mathcal{R}_d} F_{Y_d}(y|t)$$

for each $y \in \mathbb{R}$. Then, $F_{Y_d|R}(\cdot|r_d^*)$ is a well-defined cdf.

Proof. Let $l^\infty(\mathbb{R})$ denote the space of functions on $\mathbb{R}$ that are measurable and uniformly bounded and define $d_\infty(\cdot, \cdot)$ as the uniform (pseudo-)metric on $l^\infty(\mathbb{R}) \times l^\infty(\mathbb{R})$. That is,

$$l^\infty(\mathcal{Y}) \equiv \left\{ f : \sup_{y \in \mathcal{Y}} |f(y)| < \infty, f \text{ is measurable} \right\}$$

and

$$d_\infty(f, g) \equiv \sup_{y \in \mathcal{Y}} |f(y) - g(y)|.$$

We first prove the existence of $F_{Y_d|R}(\cdot|r_d^*)$. By Assumption 3.1-(ii), for any $r, \tilde{r} \in \text{Supp}(R)$, we have

$$d_\infty(F_{Y_d|R}(\cdot|r), F_{Y_d|R}(\cdot|\tilde{r})) \leq k_{d,c}|r - \tilde{r}|.$$

Thus, for any sequence $(t_n) \subset \mathcal{R}_d$ with $t_n \rightarrow r_d^*$,

$$d_\infty(F_{Y_d|R}(\cdot|t_n), F_{Y_d|R}(\cdot|t_m)) \leq k_{d,c}|t_n - t_m| \rightarrow 0$$

as $n, m \rightarrow \infty$, so $(F_{Y_d|R}(\cdot|t_n))$ is Cauchy in $(l^\infty(\mathbb{R}), d_\infty)$. Since the pseudo-metric space is complete (cf. Theorem 19.1 in Billingsley (1995)), there exists $F^* \in l^\infty(\mathbb{R})$ such that

$$d_\infty(F_{Y_d|R}(\cdot|t_n), F^*(\cdot)) \rightarrow 0,$$

and it is clear that $F_{Y_d|R}(\cdot|r_d^*) \equiv F^*(\cdot)$.

We now consider the uniqueness of $F_{Y_d|R}(\cdot|r_d^*)$. Let $(t_n) \subset \mathcal{R}_d$ and $(s_m) \subset \mathcal{R}_d$ with $t_n \rightarrow r_d^*$ and $s_m \rightarrow r_d^*$. Assume $F_{Y_d|R}(\cdot|t_n) \rightarrow F^*$ and $F_{Y_d|R}(\cdot|s_m) \rightarrow G^*$ in d_∞ . By the triangle inequality, for all $n, m \in \mathbb{N}$, we have

$$d_\infty(F^*, G^*) \leq d_\infty(F^*, F_{Y_d|R}(\cdot|t_n)) + d_\infty(F_{Y_d|R}(\cdot|t_n), F_{Y_d|R}(\cdot|s_m)) + d_\infty(F_{Y_d|R}(\cdot|s_m), G^*).$$

Taking $\limsup_{n,m \rightarrow \infty}$ yields that

$$d_\infty(F^*, G^*) \leq \limsup_{n,m \rightarrow \infty} d_\infty(F_{Y_d|R}(\cdot|t_n), F_{Y_d|R}(\cdot|s_m)).$$

By Assumption 3.1-(ii), it follows that

$$d_\infty(F_{Y_d|R}(\cdot|t_n), F_{Y_d|R}(\cdot|s_m)) \leq k_{d,c}|t_n - s_m|$$

for all $n, m \in \mathbb{N}$. Thus,

$$\limsup_{n,m \rightarrow \infty} d_\infty(F_{Y_d|R}(\cdot|t_n), F_{Y_d|R}(\cdot|s_m)) \leq \limsup_{n,m \rightarrow \infty} k_{d,c}|t_n - s_m| = 0,$$

since $t_n \rightarrow r_d^*$ and $s_m \rightarrow r_d^*$, implying that $d_\infty(F^*, G^*) = 0$.

Lastly, we prove that $F_{Y_d|R}(\cdot|r_d^*)$ is a proper distribution function. To show that $F_{Y_d|R}(\cdot|r_d^*)$ is non-decreasing, pick any $y_1 < y_2$. Since for each n , $F_{Y_d|R}(\cdot|t_n)$ is a cdf, and therefore, $F_{Y_d|R}(y_1|t_n) \leq F_{Y_d|R}(y_2|t_n)$. Since $F_{Y_d|R}(\cdot|t_n)$ uniformly converges to $F_{Y_d|R}(\cdot|r_d^*)$, we have $F_{Y_d|R}(y_1|r_d^*) \leq F_{Y_d|R}(y_2|r_d^*)$.

Take any $y \in \mathbb{R}$ and $\epsilon > 0$. By the uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d_\infty(F_{Y_d|R}(\cdot|t_N) - F_{Y_d|R}(\cdot|r_d^*)) < \frac{\epsilon}{3}.$$

Since $F_{Y_d|R}(\cdot|t_n)$ is right-continuous for all $n \in \mathbb{N}$, there exists $\delta > 0$ such that for all $h \in (0, \delta)$,

$$|F_{Y_d|R}(y+h|t_N) - F_{Y_d|R}(y|t_N)| < \frac{\epsilon}{3}.$$

Therefore, for any $h \in (0, \delta)$, we have

$$\begin{aligned}
& \left| F_{Y_d|R}(y+h|r_d^*) - F_{Y_d|R}(y|r_d^*) \right| \\
& \leq \left| F_{Y_d|R}(y+h|t_N) - F_{Y_d|R}(y+h|t_n) \right| + \left| F_{Y_d|R}(y+h|t_n) - F_{Y_d|R}(y|t_n) \right| + \left| F_{Y_d|R}(y|t_n) - F_{Y_d|R}(y|r_d^*) \right| \\
& < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon,
\end{aligned}$$

which implies that $F_{Y_d|R}(\cdot|r_d^*)$ is right-continuous.

Fix $\epsilon > 0$ and choose N such that

$$d_\infty(F_{Y_d|R}(\cdot|t_N) - F_{Y_d|R}(\cdot|r_d^*)) < \frac{\epsilon}{2}.$$

Since $F_{Y_d|R}(\cdot|t_n)$ is a cdf for all $n \in \mathbb{N}$, there exists $M > 0$ such that $F_{Y_d|R}(\cdot|t_N) < \frac{\epsilon}{2}$ for all $y \leq -M$ and $F_{Y_d|R}(\cdot|t_N) > 1 - \frac{\epsilon}{2}$ for all $y \geq M$. Then, for $y \leq -M$,

$$F_{Y_d|R}(y|r_d^*) \leq |F_{Y_d|R}(y|r_d^*) - F_{Y_d|R}(y|t_N)| + F_{Y_d|R}(y|t_N) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

meaning that $\lim_{y \rightarrow -\infty} F_{Y_d|R}(y|r_d^*) = 0$. Similarly, for $y \geq M$,

$$F_{Y_d|R}(y|r_d^*) \geq F_{Y_d|R}(y|t_N) - |F_{Y_d|R}(y|r_d^*) - F_{Y_d|R}(y|t_N)| > 1 - \frac{\epsilon}{2} - \frac{\epsilon}{2} = 1 - \epsilon,$$

and thus, $\lim_{y \rightarrow +\infty} F_{Y_d|R}(y|r_d^*) = 1$. In all, $F_{Y_d|R}(\cdot|r_d^*)$ is well-defined cdf. ■

For given $\delta \in (0, 1)$, define

$$L(u) \equiv \max\{0, u - \delta\},$$

$$U(u) \equiv \min\{1, u + \delta\},$$

as mappings from $[0, 1]$ to $[0, 1]$.

Lemma B.3. *For any $\delta \in (0, 1)$, let $\eta \in (0, \min\{\delta, \frac{1-\delta}{2}\})$ be given. Then, there exist functions $\underline{T}(\cdot; \eta)$ and $\overline{T}(\cdot; \eta)$ from $[0, 1]$ to $[0, 1]$ satisfying the following properties:*

- (T1) $\underline{T}(\cdot; \eta)$ and $\overline{T}(\cdot; \eta)$ are continuous and strictly increasing on $[0, 1]$ with $\underline{T}(0; \eta) = \overline{T}(0; \eta) = 0$ and $\underline{T}(1; \eta) = \overline{T}(1; \eta) = 1$;

(T2) for all $u \in [0, 1]$,

$$L(u) \leq \underline{T}(u; \eta) \leq U(u),$$

$$L(u) \leq \overline{T}(u; \eta) \leq U(u);$$

(T3) for any $\varepsilon \in [0, 1]$, define

$$T(u; \varepsilon, \eta) \equiv (1 - \varepsilon)\underline{T}(u; \eta) + \varepsilon\overline{T}(u; \eta)$$

for all $u \in [0, 1]$. Then, $T(\cdot; \varepsilon, \eta)$ is continuous and strictly increasing on $[0, 1]$ with $T(0; \varepsilon, \eta) = 0$ and $T(1; \varepsilon, \eta) = 1$, and

$$L(u) \leq T(u; \varepsilon, \eta) \leq U(u)$$

for all $u \in [0, 1]$;

(T4) for each fixed $u \in (0, 1)$, as $\eta \searrow 0$,

$$\underline{T}(u; \eta) \searrow L(u),$$

$$\overline{T}(u; \eta) \nearrow U(u).$$

Proof. Fix any $\eta \in \left(0, \min \left\{ \delta, \frac{(1-\delta)}{2} \right\} \right)$ and define

$$\alpha_L(\eta) \equiv \delta + \eta,$$

$$\beta_L(\eta) \equiv 1 - \eta,$$

then $0 < \delta < \alpha_L(\eta) < \beta_L(\eta) < 1$. Define $\underline{T}(\cdot; \eta) : [0, 1] \rightarrow [0, 1]$ as

$$\underline{T}(u; \eta) \equiv \begin{cases} \frac{L(\alpha_L(\eta))}{\alpha_L(\eta)} u, & 0 \leq u < \alpha_L(\eta), \\ L(u), & \alpha_L(\eta) \leq u \leq \beta_L(\eta), \\ L(\beta_L(\eta)) + \frac{1 - L(\beta_L(\eta))}{1 - \beta_L(\eta)} (u - \beta_L(\eta)), & \beta_L(\eta) \leq u \leq 1. \end{cases} \quad (9)$$

Let

$$\begin{aligned}\alpha_U(\eta) &\equiv \eta, \\ \beta_U(\eta) &\equiv 1 - \delta - \eta,\end{aligned}$$

then $0 < \alpha_U(\eta) < \beta_U(\eta) < 1 - \delta < 1$. Define $\bar{T}(\cdot; \eta) : [0, 1] \rightarrow [0, 1]$ as

$$\bar{T}(u; \eta) := \begin{cases} \frac{U(\alpha_U(\eta))}{\alpha_U(\eta)}u, & 0 \leq u < \alpha_U(\eta), \\ U(u), & \alpha_U(\eta) \leq u \leq \beta_U(\eta), \\ U(\beta_U(\eta)) + \frac{1 - U(\beta_U(\eta))}{1 - \beta_U(\eta)}(u - \beta_U(\eta)), & \beta_U(\eta) \leq u \leq 1. \end{cases} \quad (10)$$

We first prove that $\underline{T}(\cdot; \eta)$ satisfies (T1) and (T2). Note that

$$\frac{L(\alpha_L(\eta))}{\alpha_L(\eta)} = \frac{\eta}{\delta + \eta} \in (0, 1)$$

and

$$\frac{1 - L(\beta_L(\eta))}{1 - \beta_L(\eta)} = \frac{\delta + \eta}{\eta} > 1.$$

Moreover, we have $u > \delta$ and $L(u) = u - \delta$ for $u \in [\alpha_L(\eta), \beta_L(\eta)]$. Hence, each piece is linear with a positive slope and the three pieces match at $u = \alpha_L(\eta)$ and $u = \beta_L(\eta)$; thus $\underline{T}(\cdot; \eta)$ is continuous and strictly increasing on $[0, 1]$ with $\underline{T}(0; \eta) = 0$ and $\underline{T}(1; \eta) = 1$. Therefore, (T1) holds for $\underline{T}(\cdot; \eta)$.

We consider verifying that $\underline{T}(\cdot; \eta)$ satisfies (T2). For any $u \in [\alpha_L(\eta), \beta_L(\eta)]$, it is straightforward to see that $\underline{T}(u; \eta) = L(u)$ by construction, so $L(u) \leq \underline{T}(u; \eta) \leq U(u)$ holds. For $u \in [0, \alpha_L(\eta))$, we have

$$\underline{T}(u; \eta) = \frac{L(\alpha_L(\eta))}{\alpha_L(\eta)}u = \frac{\eta}{\delta + \eta}u,$$

and hence, we obtain

$$\underline{T}(u; \eta) - L(u) = \begin{cases} \frac{\eta}{\delta + \eta}u, & 0 \leq u \leq \delta, \\ \delta \frac{\alpha_L(\eta) - u}{\delta + \eta}, & \delta < u < \alpha_L(\eta), \end{cases}$$

which is nonnegative for all $u \in [0, \alpha_L(\eta))$. Therefore, it follows that $L(u) \leq \underline{T}(u; \eta)$ for all

$u \in [0, \alpha_L(\eta))$. Moreover, since $\underline{T}(\cdot; \eta)$ and $U(\cdot)$ are increasing, we have, for $u \in [0, \alpha_L(\eta)]$,

$$\underline{T}(u; \eta) \leq \underline{T}(\alpha_L(\eta); \eta) = L(\alpha_L(\eta)) = \eta \leq \delta = U(0) \leq U(u),$$

implying that $\underline{T}(u; \eta) \leq U(u)$ for $u \in [0, \alpha_L(\eta))$. Lastly, for $u \in (\beta_L(\eta), 1]$, we have that $L(u) = u - \delta$ and $L(\beta_L(\eta)) = \beta_L(\eta) - \delta$ since $u > \beta_L(\eta) > \delta$. It then follows that

$$\underline{T}(u; \eta) = L(\beta_L(\eta)) + \frac{1 - L(\beta_L(\eta))}{1 - \beta_L(\eta)}(u - \beta_L(\eta)),$$

and therefore,

$$\underline{T}(u; \eta) - L(u) = \left(\frac{1 - L(\beta_L(\eta))}{1 - \beta_L(\eta)} - 1 \right) (u - \beta_L(\eta)) \geq 0$$

since $\frac{1 - L(\beta_L(\eta))}{1 - \beta_L(\eta)} = \frac{\delta + \eta}{\eta} > 1$ and $u - \beta_L(\eta) > 0$. Thus, $\underline{T}(u; \eta) \geq L(u)$. On the other hand, since $\eta < \delta$, we have $\beta_L(\eta) = 1 - \eta > 1 - \delta$, so for $u \geq \beta_L(\eta)$,

$$U(u) = \min \{1, u + \delta\} = 1,$$

and clearly $\underline{T}(u; \eta) \leq 1 = U(u)$. Hence $\underline{T}(u; \eta) \leq U(u)$ for all $u \in (\beta_L(\eta), 1]$. In all, we obtain

$$L(u) \leq \underline{T}(u; \eta) \leq U(u)$$

for all $u \in [0, 1]$.

We now show that $\bar{T}(\cdot; \eta)$ satisfies (T1) and (T2). Note that

$$\frac{U(\alpha_U(\eta))}{\alpha_U(\eta)} = \frac{\alpha_U(\eta) + \delta}{\alpha_U(\eta)} = \frac{\eta + \delta}{\eta} > 1$$

and

$$\frac{1 - U(\beta_U(\eta))}{1 - \beta_U(\eta)} = \frac{\eta}{\delta + \eta} \in (0, 1).$$

Moreover, we have $u \leq \beta_U(\eta) \leq 1 - \delta$ for $u \in [\alpha_U(\eta), \beta_U(\eta)]$, implying that $U(u) = u + \delta$ for $u \in [\alpha_U(\eta), \beta_U(\eta)]$. Hence, each piece is linear with a positive slope, and the three pieces match at $u = \alpha_U(\eta)$ and $u = \beta_U(\eta)$; thus $\bar{T}(\cdot; \eta)$ is continuous and strictly increasing on $[0, 1]$ with $\bar{T}(0; \eta) = 0$ and $\bar{T}(1; \eta) = 1$. Therefore, (T1) holds for $\bar{T}(\cdot; \eta)$.

We verify that $\bar{T}(\cdot; \eta)$ satisfies (T2). For any $u \in [\alpha_U(\eta), \beta_U(\eta)]$, it is clear that $\bar{T}(u; \eta) = U(u)$ by construction, and thus, we have $L(u) \leq \bar{T}(u; \eta) \leq U(u)$ for all $u \in [\alpha_U(\eta), \beta_U(\eta)]$. For $u \in$

$[0, \alpha_U(\eta))$, we have $L(u) = 0$ since $u < \alpha_U(\eta) = \eta < \delta$. In addition, it follows that

$$\bar{T}(u; \eta) = \left(1 + \frac{\delta}{\eta}\right) u,$$

and therefore,

$$\bar{T}(u; \eta) - U(u) = \left(1 + \frac{\delta}{\eta}\right) u - (u + \delta) = \frac{\delta}{\eta} (u - \eta) \leq 0$$

because $0 \leq u < \eta$. Consequently, we obtain $L(u) \leq \bar{T}(u; \eta) \leq U(u)$ for $u \in [0, \alpha_U(\eta))$. Lastly, for $u \in (\beta_U(\eta), 1]$, it can be easily shown that

$$\bar{T}(u; \eta) = 1 - \eta + \frac{\eta}{\delta + \eta} (u - \beta_U(\eta)).$$

Since $\eta < \delta$, we have

$$\bar{T}(\beta_U(\eta); \eta) = 1 - \eta > 1 - \delta = L(1).$$

Because $L(\cdot)$ and $\bar{T}(\cdot; \eta)$ are nondecreasing on $[0, 1]$, it follows that, for all $u \in (\beta_U(\eta), 1]$,

$$L(u) \leq L(1) < \bar{T}(\beta_U(\eta); \eta) \leq \bar{T}(u; \eta).$$

In consequence, we have $L(u) \leq \bar{T}(u; \eta)$ for all $u \in (\beta_U(\eta), 1]$. Note that $U(u) = u + \delta$ on $u \in (\beta_U(\eta), 1]$ and $U(u) \equiv 1$ on $[1 - \delta, 1]$. Therefore, we obtain

$$U(u) - \bar{T}(u; \eta) = \begin{cases} \frac{\delta}{\delta + \eta} (\delta + \eta + u - 1), & \beta_U(\eta) < u \leq 1 - \delta, \\ \frac{\eta}{\delta + \eta} (1 - u), & 1 - \delta < u \leq 1, \end{cases}$$

and $U(u) - \bar{T}(u; \eta) \geq 0$ or all $u \in (\beta_U(\eta), 1]$. In all, we have

$$L(u) \leq \bar{T}(u; \eta) \leq U(u)$$

for all $u \in [0, 1]$. Therefore, $\underline{T}(\cdot; \eta)$ and $\bar{T}(\cdot; \eta)$ satisfy (T1) and (T2).

We now show that (T3) holds. Let $\varepsilon \in [0, 1]$ be given. For $0 \leq u_1 < u_2 \leq 1$, we have

$$\begin{aligned} T(u_2; \varepsilon, \eta) - T(u_1; \varepsilon, \eta) &= (1 - \varepsilon) \{ \underline{T}(u_2; \eta) - \underline{T}(u_1; \eta) \} \\ &\quad + \varepsilon \{ \bar{T}(u_2; \eta_d) - \bar{T}(u_1; \eta_d) \} > 0 \end{aligned}$$

since $\underline{T}(\cdot; \eta)$ and $\bar{T}(\cdot; \eta)$ are strictly increasing on $[0, 1]$ by (T1). Therefore, $T(\cdot; \varepsilon, \eta)$ is strictly

increasing on $[0, 1]$. Similarly, (T1) implies that $T(0; \varepsilon, \eta) = 0$ and $T(1; \varepsilon, \eta) = 1$. Lastly, since $T(u; \varepsilon, \eta)$ is a convex combination of $\underline{T}(u; \eta)$ and $\overline{T}(u; \eta)$, we obtain

$$L(u) \leq T(u; \varepsilon, \eta) \leq U(u)$$

for all $u \in [0, 1]$ from (T2).

We consider (T4) with $\underline{T}(u; \eta)$. Fix $u \in (0, 1)$ and suppose that $0 < u \leq \delta$. Then, for any $\eta > 0$, we have $u \leq \delta < \alpha_L(\eta)$, and thus,

$$\underline{T}(u; \eta) = \frac{L(\alpha_L(\eta))}{\alpha_L(\eta)} u = \frac{\eta}{\delta + \eta} u.$$

For any η' such that $0 < \eta' < \eta$, it then follows that

$$\underline{T}(u; \eta') - \underline{T}(u; \eta) = u \frac{\delta(\eta' - \eta)}{(\delta + \eta)(\delta + \eta')} < 0.$$

Since $\eta/(\delta + \eta) \rightarrow 0$ as $\eta \searrow 0$, we obtain $\underline{T}(u; \eta) \searrow 0 = L(u)$ as $\eta \searrow 0$ for $u \in (0, \delta]$.

Suppose that $\delta < u < 1$. Choose $\bar{\eta}(u) > 0$ such that $\delta + \bar{\eta}(u) < u < 1 - \bar{\eta}(u)$. Then, for all $0 < \eta < \bar{\eta}(u)$, we have $\alpha_L(\eta) = \delta + \eta < u < 1 - \eta = \beta_L(\eta)$, and thus,

$$\underline{T}(u; \eta) = L(u) = u - \delta$$

for all η such that $0 < \eta < \bar{\eta}(u)$ and $u \in (\delta, 1)$. In all, $\underline{T}(u; \eta) \searrow L(u)$ as $\eta \searrow 0$ for all $u \in (0, 1)$.

We now consider (T4) with $\overline{T}(u; \eta)$. Fix $u \in (0, 1)$ and suppose that $0 < u < 1 - \delta$. Choose $\bar{\eta}(u) < u < 1 - \delta - \bar{\eta}(u)$. Then, for all $0 < \eta < \bar{\eta}(u)$ we have $\alpha_U(\eta) = \eta < u < 1 - \delta - \eta = \beta_U(\eta)$, and thus,

$$\overline{T}(u; \eta) = U(u)$$

for all $0 < \eta < \bar{\eta}(u)$ and $0 < u < 1 - \delta$.

Suppose that $1 - \delta \leq u < 1$. Then, for any $\eta > 0$, we have $\beta_U(\eta) = 1 - \delta - \eta < 1 - \delta \leq u$, and thus,

$$\overline{T}(u; \eta) = U(\beta_U(\eta)) + \frac{1 - U(\beta_U(\eta))}{1 - \beta_U(\eta)} (u - \beta_U(\eta)) = 1 + \frac{\eta(u - 1)}{\delta + \eta}.$$

Take any η' such that $0 < \eta' < \eta$. Then,

$$\overline{T}(u; \eta') - \overline{T}(u; \eta) = (u - 1) \frac{\delta(\eta' - \eta)}{(\delta + \eta)(\delta + \eta')} > 0$$

since $(u - 1) < 0$ and $(\eta' - \eta) < 0$. Since $\eta(u - 1)/(\delta + \eta) \rightarrow 0$ as $\eta \searrow 0$, $\bar{T}(u; \eta) \nearrow 1 = U(u)$ as $\eta \searrow 0$ for $u \in [1 - \delta, 1)$. As a result, $\bar{T}(u; \eta) \nearrow U(u)$ as $\eta \searrow 0$ for all $u \in (0, 1)$.

This establishes that (T4) holds with $\underline{T}(u; \eta)$ and $\bar{T}(u; \eta)$ defined in (9) and (10). $\blacksquare$

Proof of Theorem 3.3

Proof. Proof of Claim (a): We show that $F_{Y_d|R}(\cdot \mid r_{d'}) \in \mathcal{F}_{Y_d|R}^{k_{d,c}}$ for given $d \in \{0, 1\}$ and $r_{d'} \in \mathcal{R}_{d'}$, where $d' = 1 - d$. For any $\tilde{r} \in \mathcal{R}_d$ and $y \in \mathbb{R}$, we have

$$|F_{Y_d|R}(y \mid r_{d'}) - F_{Y_d|R}(y \mid \tilde{r})| \leq k_{d,c}|r_{d'} - \tilde{r}|$$

by Assumption 3.1. It follows that

$$F_{Y_d|R}(y \mid \tilde{r}) - k_{d,c}|r_{d'} - \tilde{r}| \leq F_{Y_d|R}(y \mid r_{d'}) \leq F_{Y_d|R}(y \mid \tilde{r}) + k_{d,c}|r_{d'} - \tilde{r}|.$$

Taking the supremum and infimum over $\tilde{r} \in \mathcal{R}_d$ yields

$$\sup_{\tilde{r} \in \mathcal{R}_d} \left\{ F_{Y_d|R}(y \mid \tilde{r}) - k_{d,c}|r_{d'} - \tilde{r}| \right\} \leq F_{Y_d|R}(y \mid r_{d'}) \leq \inf_{\tilde{r} \in \mathcal{R}_d} \left\{ F_{Y_d|R}(y \mid \tilde{r}) + k_{d,c}|r_{d'} - \tilde{r}| \right\}. \quad (11)$$

Since $F_{Y_d|R}(y \mid r_d^*)$ is well defined by Lemma B.2 for each $d \in \{0, 1\}$, we have

$$\sup_{\tilde{r} \in \mathcal{R}_d} \left\{ F_{Y_d|R}(y \mid \tilde{r}) - k_{d,c}|r_{d'} - \tilde{r}| \right\} = F_{Y_d|R}(y \mid r_d^*) - k_{d,c}|r_{d'} - r_d^*|$$

and

$$\inf_{\tilde{r} \in \mathcal{R}_d} \left\{ F_{Y_d|R}(y \mid \tilde{r}) + k_{d,c}|r_{d'} - \tilde{r}| \right\} = F_{Y_d|R}(y \mid r_d^*) + k_{d,c}|r_{d'} - r_d^*|,$$

by Theorem 3.2. Invoking the logical bounds on the cdf, we complete the proof.

Proof of Claim (b): For $d \in \{0, 1\}$, let $d' \equiv 1 - d$ and $\delta_d(r_{d'}) \equiv k_{d,c}|r_{d'} - r_d^*|$. If $\delta_d(r_{d'}) = 0$, then

$$\underline{F}_{Y_d|R}^{k_{d,c}}(\cdot \mid r_{d'}) = \bar{F}_{Y_d|R}^{k_{d,c}}(\cdot \mid r_{d'}) = F_{Y_d|R}(\cdot \mid r_d^*),$$

and no further construction is required. If $\delta_d(r_{d'}) \geq 1$, then

$$\underline{F}_{Y_d|R}^{k_{d,c}}(\cdot \mid r_{d'}) \equiv 0, \bar{F}_{Y_d|R}^{k_{d,c}}(\cdot \mid r_{d'}) \equiv 1,$$

and there is nothing to prove. Therefore, we focus on the case that $\delta_d(r_{d'}) \in (0, 1)$.

Note that

$$\underline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'}) = L_d(F_{Y_d|R}(y \mid r_d^*))$$

and

$$\overline{F}_{Y_d|R}^{k_d}(y \mid r_{d'}) = U_d(F_{Y_d|R}(y \mid r_d^*)),$$

where $L_d, U_d : [0, 1] \rightarrow [0, 1]$ are defined by

$$L_d(u) \equiv \max\{0, u - \delta_d(r_{d'})\},$$

$$U_d(u) \equiv \min\{1, u + \delta_d(r_{d'})\}.$$

Let $\eta_d \in (0, \min\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\})$ be given and define $\underline{T}^{k_d,c}(\cdot; \eta_d)$ and $\overline{T}^{k_d,c}(\cdot; \eta_d)$ on $[0, 1]$

as

$$\underline{T}^{k_d,c}(u; \eta_d) \equiv \begin{cases} \frac{L_d(\alpha_L(\eta_d))}{\alpha_L(\eta_d)}u, & 0 \leq u < \alpha_L(\eta_d), \\ L_d(u), & \alpha_L(\eta_d) \leq u \leq \beta_L(\eta_d), \\ L_d(\beta_L(\eta_d)) + \frac{1 - L_d(\beta_L(\eta_d))}{1 - \beta_L(\eta_d)}(u - \beta_L(\eta_d)), & \beta_L(\eta_d) \leq u \leq 1, \end{cases}$$

and

$$\overline{T}^{k_d,c}(u; \eta_d) \equiv \begin{cases} \frac{U_d(\alpha_U(\eta_d))}{\alpha_U(\eta_d)}u, & 0 \leq u < \alpha_U(\eta_d), \\ U_d(u), & \alpha_U(\eta_d) \leq u \leq \beta_U(\eta_d), \\ U_d(\beta_U(\eta_d)) + \frac{1 - U_d(\beta_U(\eta_d))}{1 - \beta_U(\eta_d)}(u - \beta_U(\eta_d)), & \beta_U(\eta_d) \leq u \leq 1, \end{cases}$$

where $\alpha_L(\eta_d) \equiv \delta_d(r_{d'}) + \eta_d$, $\beta_L(\eta_d) \equiv 1 - \eta_d$, $\alpha_U(\eta_d) \equiv \eta_d$, and $\beta_U(\eta_d) := 1 - \delta_d - \eta_d$. Then,

Lemma B.3 implies that $\underline{T}^{k_d,c}(\cdot; \eta_d)$ and $\overline{T}^{k_d,c}(\cdot; \eta_d)$ satisfy the following properties:

(Td1) $\underline{T}^{k_d,c}(\cdot; \eta_d)$ and $\overline{T}^{k_d,c}(\cdot; \eta_d)$ are continuous and strictly increasing on $[0, 1]$ with $\underline{T}^{k_d,c}(0; \eta_d) = \overline{T}^{k_d,c}(0; \eta_d) = 0$ and $\underline{T}^{k_d,c}(1; \eta_d) = \overline{T}^{k_d,c}(1; \eta_d) = 1$;

(Td2) for all $u \in [0, 1]$,

$$L_d(u) \leq \underline{T}^{k_d,c}(u; \eta_d) \leq U_d(u),$$

$$L_d(u) \leq \overline{T}^{k_d,c}(u; \eta_d) \leq U_d(u);$$

(Td3) for any $\varepsilon_d \in [0, 1]$, define

$$T^{k_d, c}(u; \varepsilon_d, \eta_d) \equiv (1 - \varepsilon_d) \underline{T}^{k_d, c}(u; \eta_d) + \varepsilon_d \overline{T}^{k_d, c}(u; \eta_d) \quad (12)$$

for all $u \in [0, 1]$. Then, $T^{k_d, c}(\cdot; \varepsilon_d, \eta_d)$ is continuous and strictly increasing on $[0, 1]$ with $T^{k_d, c}(0; \varepsilon_d, \eta_d) = 0$ and $T^{k_d, c}(1; \varepsilon_d, \eta_d) = 1$, and

$$L_d(u) \leq T^{k_d, c}(u; \varepsilon_d, \eta_d) \leq U_d(u) \quad \text{for all } u \in [0, 1];$$

(Td4) for each fixed $u \in (0, 1)$, as $\eta_d \searrow 0$,

$$\underline{T}^{k_d, c}(u; \eta_d) \searrow L_d(u)$$

and

$$\overline{T}^{k_d, c}(u; \eta_d) \nearrow U_d(u).$$

Let $\epsilon_d \in [0, 1]$ be given and define

$$F_{Y_d|R}^{k_d, c}(y \mid r_{d'}; \varepsilon_d, \eta_d) \equiv T^{k_d, c}(F_{Y_d|R}(y \mid r_d^*); \varepsilon_d, \eta_d). \quad (13)$$

We will proceed with showing that (1) for each $\epsilon_d \in [0, 1]$ and $\eta_d \in \left(0, \min \left\{ \delta_d(r_{d'}), \frac{1 - \delta_d(r_{d'})}{2} \right\} \right)$, $F_{Y_d|R}^{k_d, c}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is a valid cdf and lies in the cdf band at $R = r_d^*$; (2) for each $y \in \mathbb{R}$ and $\eta_d \in \left(0, \min \left\{ \delta_d(r_{d'}), \frac{1 - \delta_d(r_{d'})}{2} \right\} \right)$, $F_{Y_d|R}^{k_d, c}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous in ϵ_d on $[0, 1]$.

Note that $F_{Y_d|R}(\cdot \mid r_d^*)$ is a cdf by Lemma B.2. It also follows from (Td3) that for any fixed (ε_d, η_d) the map $T^{k_d, c}(\cdot; \varepsilon_d, \eta_d)$ is continuous and strictly increasing from $[0, 1]$ to $[0, 1]$ such that $T^{k_d, c}(0; \varepsilon_d, \eta_d) = 0$ and $T^{k_d, c}(1; \varepsilon_d, \eta_d) = 1$. Therefore, $F_{Y_d|R}^{k_d, c}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d) = T^{k_d, c}(F_{Y_d|R}(\cdot \mid r_d^*); \varepsilon_d, \eta_d)$ is a valid cdf. Moreover, by (Td3),

$$L_d(u) \leq T^{k_d, c}(u; \varepsilon_d, \eta_d) \leq U_d(u)$$

for all $u \in [0, 1]$. Therefore, substituting $u = F_{Y_d|R}(y \mid r_d^*)$ yields that $\underline{F}_{Y_d|R}^{k_d, c}(y \mid r_{d'}) = L_d(F_{Y_d|R}(y \mid r_d^*))$ and $\overline{F}_{Y_d|R}^{k_d, c}(y \mid r_{d'}) = U_d(F_{Y_d|R}(y \mid r_d^*))$. Combining these together, we obtain that, for all $y \in \mathbb{R}$,

$$\underline{F}_{Y_d|R}^{k_d, c}(y \mid r_{d'}) \leq F_{Y_d|R}^{k_d, c}(y \mid r_{d'}; \varepsilon_d, \eta_d) \leq \overline{F}_{Y_d|R}^{k_d, c}(y \mid r_{d'}),$$

implying that $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}_{Y_d|r_{d'}}^{k_{d,c}}$.

We now show that $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$ satisfies (i), (ii) and (iii) in Claim (b) of the theorem.

Let $F_R(\cdot)$ denote the distribution function of R . Since $F_R(\cdot)$ is identified from data, it is enough to show that $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$ for $d \in \{0, 1\}$ can be jointly obtained. To this end, we construct conditional cdfs $\tilde{F}_{Y_d|R}(\cdot | r)$ for all $r \in \text{Supp}(R)$ and $d \in \{0, 1\}$ such that:

(S1) they coincide with $F_{Y_d|R}^{k_d}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$ at $r_{d'} \in \mathcal{R}_{d'}$;

(S2) for each $r \in \text{Supp}(R)$ and $d \in \{0, 1\}$, $\tilde{F}_{Y_d|R}(\cdot | r)$ is a cdf;

(S3) they satisfy Assumption 3.1.

Fix $d \in \{0, 1\}$ and let $d' \equiv 1 - d$. By construction, $\mathcal{R}_d \cap \mathcal{R}_{d'} = \emptyset$ and $\mathcal{R}_d \cup \mathcal{R}_{d'} = \text{Supp}(R)$. Let r_d^* denote the boundary point on side d ; by Lemma A.1, $F_{Y_d|R}(\cdot | r_d^*)$ is well-defined and a c.d.f. Let $\Delta_d := |r_{d'} - r_d^*|$ and $\delta_d \equiv k_{d,c}\Delta_d$ as in Part 2. Define the weight function $\tau_d : \mathcal{R}_{d'} \rightarrow [0, 1]$ by

$$\tau_d(r) := \min\left\{\frac{|r - r_d^*|}{\Delta_d}, 1\right\}$$

for $r_d \in \mathcal{R}_d$. Fix $\varepsilon_d \in [0, 1]$ and $\eta_d \in (0, \min\{\delta_d, (1 - \delta_d)/2\})$. For $r \in \text{Supp}(R)$ and $y \in \mathbb{R}$, define

$$\tilde{F}_{Y_d|R}(y | r; \varepsilon_d, \eta_d) \equiv \begin{cases} F_{Y|R,D=d}(y | r), & r \in \mathcal{R}_d, \\ (1 - \tau_d(r)) F_{Y_d|R}(y | r_d^*) + \tau_d(r) F_{Y_d|R}^{k_d}(y | r_{d'}; \varepsilon_d, \eta_d), & r \in \mathcal{R}_{d'}. \end{cases} \quad (14)$$

Since $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}_{Y_d|r_{d'}}^{k_{d,c}}$, we have

$$\sup_{y \in \mathbb{R}} |F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y | r_d^*)| \leq \delta_d = k_d \Delta_d. \quad (15)$$

We now verify properties 1–3 above for this choice. For convenience, write $\tilde{F}_{Y_d|R}(\cdot | r) := \tilde{F}_{Y_d|R}(\cdot | r; \varepsilon_d, \eta_d)$ in the remainder of this part.

1. Since $r_{d'} \in \mathcal{R}_{d'}$ and $\tau_d(r_{d'}) = 1$, we have $\tilde{F}_{Y_d|R}(\cdot | r_{d'}) = F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$.
2. If $r \in \mathcal{R}_d$, then by (14) we have $\tilde{F}_{Y_d|R}(\cdot | r) = F_{Y|R,D=d}(\cdot | r)$, a conditional distribution in y and thus a c.d.f. If $r \in \mathcal{R}_{d'}$, then $\tilde{F}_{Y_d|R}(\cdot | r)$ is a convex combination (with the weight $\tau_d(r) \in [0, 1]$) of the cdf $F_{Y_d|R}(\cdot | r_d^*)$ and $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$. Convex combinations preserve monotonicity, right-continuity, and the limits 0 and 1; hence $\tilde{F}_{Y_d|R}(\cdot | r)$ is a cdf for every $r \in \text{Supp}(R)$.

3. By construction, on $\mathcal{R}_d$ we have $\tilde{F}_{Y_d|R}(\cdot | r) = F_{Y|R, D=d}(\cdot | r)$, so Assumption DI holds. For Assumption 3.1, we consider three cases.

Case 1: $r, r' \in \mathcal{R}_d$. For any $y \in \mathbb{R}$,

$$\begin{aligned} |\tilde{F}_{Y_d|R}(y | r) - \tilde{F}_{Y_d|R}(y | r')| &= |F_{Y|R, D=d}(y | r) - F_{Y|R, D=d}(y | r')| \\ &= |F_{Y_d|R}(y | r) - F_{Y_d|R}(y | r')| \\ &\leq k_{d,c}|r - r'|, \end{aligned}$$

where the last inequality follows from Assumption DS for $F_{Y_d|R}$.

Case 2: $r, r' \in \mathcal{R}_{d'}$. For any $y \in \mathbb{R}$,

$$\begin{aligned} |\tilde{F}_{Y_d|R}(y | r) - \tilde{F}_{Y_d|R}(y | r')| &= |\tau_d(r) - \tau_d(r')| \cdot |F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y | r_d^*)| \\ &\leq \frac{|r - r'|}{\Delta_d} k_{d,c} \Delta_d = k_{d,c}|r - r'|, \end{aligned}$$

where the inequality uses $|\tau_d(r) - \tau_d(r')| \leq |r - r'|/\Delta_d$ and (15).

Case 3: $r \in \mathcal{R}_d, r' \in \mathcal{R}_{d'}$. For any $y \in \mathbb{R}$,

$$\begin{aligned} |\tilde{F}_{Y_d|R}(y | r') - \tilde{F}_{Y_d|R}(y | r)| &\leq |\tilde{F}_{Y_d|R}(y | r') - F_{Y_d|R}(y | r_d^*)| + |F_{Y_d|R}(y | r_d^*) - \tilde{F}_{Y_d|R}(y | r)| \\ &= \tau_d(r') \cdot |F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y | r_d^*)| + |F_{Y_d|R}(y | r_d^*) - F_{Y|R, D=d}(y | r)| \\ &= \tau_d(r') |F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y | r_d^*)| + |F_{Y_d|R}(y | r_d^*) - F_{Y_d|R}(y | r)| \\ &\leq \frac{|r' - r_d^*|}{\Delta_d} k_{d,c} \Delta_d + k_{d,c}|r - r_d^*| \\ &= k_{d,c}|r' - r|, \end{aligned}$$

where the first line is the triangle inequality, the second line uses (14), the third line replaces $F_{Y|R, D=d}$ by $F_{Y_d|R}$, the fourth line uses $|\tau_d(r')| \leq |r' - r_d^*|/\Delta_d$, (15), and Assumption 3.1 for $F_{Y_d|R}$, and the last equality uses that r_d^* lies between r and r' .

We turn to prove that $F_{Y_d|R}^{k_{d,c}}(\cdot | r_{d'}; \varepsilon_d, \eta_d)$ satisfies (ii). Fix $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y | r_d^*) < 1$ and set $u_d^* \equiv F_{Y_d|R}(y | r_d^*) \in (0, 1)$. Note that

$$F_{Y_d|R}^{k_{d,c}}(y | r_{d'}; \varepsilon_d, \eta_d) = T^{k_{d,c}}(u_d^*; \varepsilon_d, \eta_d)$$

and

$$\begin{aligned} L_d(u^*) &= \underline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'}), \\ U_d(u^*) &= \overline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'}). \end{aligned}$$

Applying the monotone convergence leads to that, as $\eta_d \searrow 0$,

$$F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; 0, \eta_d) = \underline{T}^{k_d,c}(u^*; \eta_d) \searrow L_d(u^*) = \underline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'})$$

and

$$F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; 1, \eta_d) = \overline{T}^{k_d,c}(u^*; \eta_d) \nearrow U_d(u^*) = \overline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'}).$$

Lastly, observe that $T^{k_d,c}(u; \varepsilon_d, \eta_d) = (1 - \varepsilon_d)\underline{T}^{k_d,c}(u; \eta_d) + \varepsilon_d\overline{T}^{k_d,c}(u; \eta_d)$ for fixed $y \in \mathbb{R}$ and η_d . Therefore,

$$\begin{aligned} F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; \varepsilon_d, \eta_d) &= T^{k_d,c}(F_{Y_d|R}(y \mid r_d^*); \varepsilon_d, \eta_d) \\ &= (1 - \varepsilon_d)\underline{T}^{k_d,c}(F_{Y_d|R}(y \mid r_d^*); \eta_d) + \varepsilon_d\overline{T}^{k_d,c}(F_{Y_d|R}(y \mid r_d^*); \eta_d). \end{aligned}$$

Since $(1 - \varepsilon_d)\underline{T}^{k_d,c}(F_{Y_d|R}(y \mid r_d^*); \eta_d) + \varepsilon_d\overline{T}^{k_d,c}(F_{Y_d|R}(y \mid r_d^*); \eta_d)$ is a linear function in ε_d , $F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous in ε_d on $[0, 1]$.

Proof of Claim (c): We apply the above results to obtain pointwise bounds at the target point $r_{d'}$. Fix $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid r_d^*) < 1$ and let $e \in \left(\underline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'}), \overline{F}_{Y_d|R}^{k_d,c}(y \mid r_{d'})\right)$. By (ii) in Claim (b), there exists $\eta_d^* > 0$ such that $e \in \left[F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; 0, \eta_d^*), F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; 1, \eta_d^*)\right]$. Since the map $\varepsilon_d \mapsto F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; \varepsilon_d, \eta_d^*)$ is continuous on $[0, 1]$ by (iii) in Claim (b), there exists $\varepsilon_d^* \in [0, 1]$ such that $e = F_{Y_d|R}^{k_d,c}(y \mid r_{d'}; \varepsilon_d^*, \eta_d^*)$ by the intermediate value theorem. Thus, the value e is attainable. $\blacksquare$

B.4 Proof of Theorem 3.5

Let $\tau \in (0, 1)$ and $\epsilon \in [0, 1]$ be given. Fix $d \in \{0, 1\}$ and $\eta_d \in \left(0, \min\left\{\delta_d(r_{d'}), \frac{1 - \delta_d(r_{d'})}{2}\right\}\right)$ with $d' = 1 - d$ and $\delta_d(r_{d'}) \equiv k_{d,c}|r_{d'} - r_d^*|$. Define $(T^{k_d,c})^{-1}(\tau, \epsilon, \eta_d)$ as the value u such that $T^{k_d,c}(u, \epsilon, \eta_d) = \tau$, where the definition of $T^{k_d,c}(u, \epsilon, \eta_d)$ is in equation (12).

Lemma B.4. *Suppose that Assumptions 3.1 and 3.2 hold. Fix $d \in \{0, 1\}$ and $\eta_d \in \left(0, \min\left\{\delta_d(r_{d'}), \frac{1 - \delta_d(r_{d'})}{2}\right\}\right)$. Then, for each $\tau \in (0, 1)$, the map $\epsilon \mapsto (T^{k_d,c})^{-1}(\tau, \epsilon, \eta_d)$ is continuous on $[0, 1]$.*

Proof. For simplicity of notation, let $T^{-1}(\tau; \varepsilon) = (T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$ for a given value of η_d . Note that for each $\tau \in (0, 1)$, the value of $(T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$ exists and is unique since $T^{k_{d,c}}(\cdot, \varepsilon, \eta_d)$ is continuous and strictly increasing from the proof of Theorem 3.3.

Let (ε_n) be any sequence in $[0, 1]$ with $\varepsilon_n \rightarrow \varepsilon$ as $n \rightarrow \infty$. For each n , define

$$u_n \equiv (T^{k_{d,c}})^{-1}(\tau, \varepsilon_n, \eta_d) \in [0, 1],$$

so that, by definition of $(T^{k_{d,c}})^{-1}$,

$$T^{k_{d,c}}(u_n, \varepsilon_n, \eta_d) = \tau$$

for all $n \in \mathbb{N}$. Since (u_n) is a sequence in the compact set $[0, 1]$, there exists a subsequence (u_{n_m}) such that $u_{n_m} \rightarrow u^*$ for some $u^* \in [0, 1]$ as $m \rightarrow \infty$. Because $\varepsilon_{n_m} \rightarrow \varepsilon$ and $T^{k_{d,c}}(\cdot, \cdot, \eta_d)$ is continuous on $[0, 1] \times [0, 1]$, we obtain

$$T^{k_{d,c}}(u^*, \varepsilon, \eta_d) = \lim_{k \rightarrow \infty} T^{k_{d,c}}(u_{n_m}, \varepsilon_{n_m}, \eta_d) = \lim_{m \rightarrow \infty} \tau = \tau.$$

For this fixed ε , the map $u \mapsto T^{k_{d,c}}(u, \varepsilon, \eta_d)$ is strictly increasing and continuous on $[0, 1]$, with $T^{k_{d,c}}(0, \varepsilon, \eta_d) = 0$ and $T^{k_{d,c}}(1, \varepsilon, \eta_d) = 1$ from the proof of Theorem 3.3. Hence, the equation $T^{k_{d,c}}(u, \varepsilon, \eta_d) = \tau$ has a unique solution in $[0, 1]$, namely $u^* = (T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$. Thus, every convergent subsequence of (u_n) has the limit $(T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$. Hence (u_n) converges to $(T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$, which implies that

$$(T^{k_{d,c}})^{-1}(\tau, \varepsilon_n, \eta_d) \rightarrow (T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$$

as $n \rightarrow \infty$. Since (ε_n) was arbitrary, the map $\varepsilon \mapsto (T^{k_{d,c}})^{-1}(\tau, \varepsilon, \eta_d)$ is continuous on $[0, 1]$. ■

Let

$$Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d) \equiv \inf \left\{ y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; \varepsilon_d, \eta_d) \geq \tau \right\},$$

where the definition of $F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is in (13).

Lemma B.5. *Suppose that Assumptions 3.1 and 3.2 hold. Then, the map $\varepsilon_d \mapsto Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous on $[0, 1]$.*

Pick any $\tau \in (0, 1)$ and any sequence (η_n) with $0 < \eta_n < \min\{\delta_d(r_{d'}), (1 - \delta_d(r_{d'}))/2\}$ for all n and $\eta_n \searrow 0$. Note that, from the proof of Theorem 3.3 that, for each $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid$

$r_d^*) < 1$, we have

$$\begin{aligned} F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 1, \eta_d) &\nearrow \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}), \\ F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 0, \eta_d) &\searrow \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}). \end{aligned}$$

as $\eta_d \searrow 0$. Therefore, the sequence of sets $\mathcal{Y}_{1n} \equiv \{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 1, \eta_n) \geq \tau\}$ is monotonically increasing. In consequence, the sequence $y_{1n} \equiv \inf \mathcal{Y}_{1n}$ is monotonically decreasing and converges to

$$\inf \{y \in \mathbb{R} : \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\} = \underline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}),$$

which may be $-\infty$. Next, we observe that the sequence of sets $\mathcal{Y}_{0n} \equiv \{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 0, \eta_n) \geq \tau\}$ is monotonically decreasing, implying that the sequence $y_{0n} \equiv \inf \mathcal{Y}_{0n}$ is monotonically increasing and converges to

$$\inf \{y \in \mathbb{R} : \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\} = \overline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}),$$

which may be $+\infty$. Note that

$$\begin{aligned} Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d) &= \inf \{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; \varepsilon_d, \eta_d) \geq \tau\} \\ &= \inf \{y \in \mathbb{R} : T^{k_{d,c}}(F_{Y_d|R}(y \mid r_d^*); \varepsilon_d, \eta_d) \geq \tau\} \\ &= \inf \{y \in \mathbb{R} : F_{Y_d|R}(y \mid r_d^*) \geq (T^{k_{d,c}})^{-1}(\tau, \varepsilon_d, \eta_d)\} \\ &= Q_{Y_d|R}((T^{k_{d,c}})^{-1}(\tau, \varepsilon_d, \eta_d) \mid r_d^*). \end{aligned}$$

By Assumption 3.2, $Q_{Y_d|R}(\cdot \mid r_d^*)$ is continuous on $(0, 1)$. By Lemma B.4, for each $\tau \in (0, 1)$ and η_d , the map $\varepsilon_d \mapsto (T^{k_{d,c}})^{-1}(\tau, \varepsilon_d, \eta_d)$ is continuous on $[0, 1]$, and therefore, the map $\varepsilon_d \mapsto Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous on $[0, 1]$.

Proof of Theorem 3.5

Proof. Let $d \in \{0, 1\}$ and $r_{d'} \in \mathcal{R}_{d'}$. Recall that by Theorem 3.3, we have

$$\underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \leq F_{Y_d|R}(y \mid r_{d'}) \leq \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}),$$

where

$$\begin{aligned} \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) &\equiv \max \{F_{Y_d|R}(y \mid r_d^*) - k_{d,c}|r_{d'} - r_d^*|, 0\}, \\ \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) &\equiv \min \{F_{Y_d|R}(y \mid r_d^*) + k_{d,c}|r_{d'} - r_d^*|, 1\}. \end{aligned}$$

Thus, for any $\tau \in (0, 1)$,

$$\left\{y : \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} \subseteq \left\{y : F_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} \subseteq \left\{y : \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\}.$$

Taking infima yields

$$\inf \left\{y : \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} \leq \inf \left\{y : F_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} = Q_{Y_d|R}(\tau \mid r_d) \leq \inf \left\{y : \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\}.$$

We first consider the lower bound. Note that

$$\begin{aligned} \inf \left\{y : \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} &= \inf \left\{y : \min \{1, F_{Y_d|R}(y \mid r_d^*) + \delta_d(r_{d'})\} \geq \tau\right\} \\ &= \inf \left\{y : F_{Y_d|R}(y \mid r_d^*) + \delta_d(r_{d'}) \geq \tau\right\} \\ &= \inf \left\{y : F_{Y_d|R}(y \mid r_d^*) \geq \tau - \delta_d(r_{d'})\right\}, \end{aligned}$$

where the second line follows from $\tau < 1$ and the fact that for any $a \in \mathbb{R}$, $\min \{1, a\} \geq \tau \iff (1 \geq \tau \text{ and } a \geq \tau) \iff a \geq \tau$.

From the last equality, we evaluate the infimum as follows. If $\tau \leq \delta_d(r_{d'})$, then $\tau - \delta_d(r_{d'}) \leq 0$ and

$$\left\{y : F_{Y_d|R}(y \mid r_d^*) \geq \tau - \delta_d(r_{d'})\right\} = \left\{y : F_{Y_d|R}(y \mid r_d^*) \geq 0\right\},$$

so the infimum equals $-\infty$. If $\tau > \delta_d(r_{d'})$, then $0 < \tau - \delta_d(r_{d'}) < 1$ and, by the definition of the quantile,

$$\inf \left\{y : F_{Y_d|R}(y \mid r_d^*) \geq \tau - \delta_d(r_{d'})\right\} = Q_{Y|R, D=d}(\tau - \delta_d(r_{d'}) \mid r_d^*).$$

Therefore, by the definition of $\underline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'})$,

$$\underline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}) \leq Q_{Y_d|R}(\tau \mid r_{d'}).$$

We now consider the upper bound. Recall that

$$\begin{aligned} \inf \left\{y : \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} &= \inf \left\{y : \max \{0, F_{Y_d|R}(y \mid r_d^*) - \delta_d(r_{d'})\} \geq \tau\right\} \\ &= \inf \left\{y : F_{Y_d|R}(y \mid r_d^*) - \delta_d(r_{d'}) \geq \tau\right\} \\ &= \inf \left\{y : F_{Y_d|R}(y \mid r_d^*) \geq \tau + \delta_d(r_{d'})\right\}, \end{aligned}$$

where the second line follows from $\tau > 0$ and the fact that for any $a \in \mathbb{R}$, $\max \{0, a\} \geq \tau \iff$

$$(0 \geq \tau) \text{ or } (a \geq \tau) \iff a \geq \tau.$$

From the last equality, we evaluate the infimum as follows. If $\tau + \delta_d(r_{d'}) > 1$, then

$$\{y : y : F_{Y_d|R}(y \mid r_d^*) \geq \tau + \delta_d(r_{d'})\} = \emptyset,$$

and thus the upper bound is $+\infty$. If $\tau + \delta_d(r_{d'}) \leq 1$, then $0 < \tau + \delta_d(r_{d'}) \leq 1$ and, by the definition of the quantile,

$$\inf \{y : F_{Y_d|R}(y \mid r_d^*) \geq \tau + \delta_d(r_{d'})\} = Q_{Y|R, D=d}(\tau + \delta_d(r_{d'}) \mid r_d^*).$$

Therefore, by the definition of $\overline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'})$,

$$Q_{Y_d|R}(\tau \mid r_{d'}) \leq \overline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}).$$

For $d \in \{0, 1\}$, $r \in \text{Supp}(R)$, $\tau \in (0, 1)$ and (ε_d, η_d) , define the left-quantile of the extended cdf by

$$\tilde{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r; \varepsilon_d, \eta_d) \equiv \inf \left\{ y \in \mathbb{R} : \tilde{F}_{Y_d|R}(y \mid r; \varepsilon_d, \eta_d) \geq \tau \right\},$$

where $\tilde{F}_{Y_d|R}(y \mid r; \varepsilon_d, \eta_d)$ is defined in (14).

We distinguish two cases.

Case 1) $r \in \mathcal{R}_1$: By the definition of the extension,

$$\begin{aligned} & \tilde{Q}_{Y_1|R}^{k_{1,c}}(\tau \mid r; \varepsilon_1, \eta_1) - \tilde{Q}_{Y_0|R}^{k_{0,c}}(\tau \mid r; \varepsilon_0, \eta_0) \\ &= \inf \left\{ y \in \mathbb{R} : F_{Y|R, D=1}(y \mid r) \geq \tau \right\} - \inf \left\{ y \in \mathbb{R} : F_{Y_0|R}^{k_{0,c}}(y \mid r; \varepsilon_0, \eta_0) \geq \tau \right\} \\ &= Q_{Y|R, D=1}(\tau \mid r) - Q_{Y_0|R}^{k_{0,c}}(\tau \mid r; \varepsilon_0, \eta_0) \end{aligned} \tag{16}$$

Let $e \in \left(\underline{\text{CQTE}}^{k_{0,c}}(\tau \mid r), \overline{\text{CQTE}}^{k_{0,c}}(\tau \mid r) \right)$. By the monotone approximation, there exists $\eta_0^* > 0$ such that e is above the value of equation (16) evaluated at $(\varepsilon_0, \eta_0) = (0, \eta_0^*)$ and below the value of the equation evaluated at $(\varepsilon_0, \eta_0) = (1, \eta_0^*)$. Next, note that $Q_{Y_0|R}^{k_{0,c}}(\tau \mid r; \varepsilon_0, \eta_0)$ is continuous in ε_0 on $[0, 1]$ by Lemma B.5. Thus, by the intermediate value theorem, there exists $\varepsilon_0^* \in [0, 1]$ such that equation (16) evaluated at $(\varepsilon_0, \eta_0) = (\varepsilon_0^*, \eta_0^*)$ yields e .

Case 2) $r \in \mathcal{R}_0$: By the definition of the extension,

$$\begin{aligned}
& \tilde{Q}_{Y_1|R}^{k_1,c}(\tau | r; \varepsilon_1, \eta_1) - \tilde{Q}_{Y_0|R}^{k_0,c}(\tau | r; \varepsilon_0, \eta_0) \\
&= \inf \left\{ y \in \mathbb{R} : F_{Y_1|R}^{k_1,c}(y | r; \varepsilon_1, \eta_1) \geq \tau \right\} - \inf \left\{ y \in \mathbb{R} : F_{Y|R,D=0}(y | r) \geq \tau \right\} \\
&= Q_{Y_1|R}^{k_1,c}(\tau | r; \varepsilon_1, \eta_1) - Q_{Y|R,D=0}(\tau | r).
\end{aligned} \tag{17}$$

Let $e \in \left(\underline{\text{CQTE}}^{k_1,c}(\tau | r), \overline{\text{CQTE}}^{k_1,c}(\tau | r) \right)$. By the monotone approximation, there exists $\eta_1^* > 0$ such that e is above the value of equation (17) evaluated at $(\varepsilon_1, \eta_1) = (1, \eta_1^*)$ and below the value of the equation evaluated at $(\varepsilon_1, \eta_1) = (0, \eta_1^*)$. Next, note that $Q_{Y_1|R}^{k_1,c}(\tau | r; \varepsilon_1, \eta_1)$ is continuous in ε_1 on $[0, 1]$ Lemma B.5. Thus, by the intermediate value theorem, there exists $\varepsilon_1^* \in [0, 1]$ such that equation (17) evaluated at $(\varepsilon_1, \eta_1) = (\varepsilon_1^*, \eta_1^*)$ yields e .

In all, every value in the interior of $\left[\underline{\text{CQTE}}^{k_c}(\tau | r), \overline{\text{CQTE}}^{k_c}(\tau | r) \right]$ is attainable, so the interior of the identified set is sharp. ■

C Identification of the Means of Counterfactual Outcomes

In this section, we investigate the identification of the mean of counterfactual outcomes without considering identification of the distribution functions. We assume that $E[|Y_d|] < \infty$ and define $g_d(r) \equiv E[Y_d | R = r]$.

Assumption C.1. *The following conditions hold:*

- (i) *For each $d \in \{0, 1\}$, $E[Y_d | R = r_d]$ is identified for any $r_d \in \mathcal{R}_d$.*
- (ii) *Let $r, \tilde{r} \in \text{Supp}(R)$ be given. There exist $k_{0,m}, k_{1,m} \geq 0$ such that*

$$\left| E[Y_1 | R = r] - E[Y_1 | R = \tilde{r}] \right| \leq k_{1,m} |r - \tilde{r}|$$

and

$$\left| E[Y_0 | R = r] - E[Y_0 | R = \tilde{r}] \right| \leq k_{0,m} |r - \tilde{r}|.$$

Assumption C.1-(i) can be viewed as a special case of Assumption 2.1. Specifically, Assumption C.1-(i) holds when Assumption 2.1 is satisfied with $h(x) = x$. Assumption C.1-(ii) imposes a global Lipschitz restriction on $g_d(r)$. The following theorem formalizes the corresponding extrapolation results.

Theorem C.1. *Suppose that Assumption C.1 holds. For any given $r_d \in \mathcal{R}_d$ with $d \in \{0, 1\}$, we have*

$$\begin{aligned} E[Y_1|R = r_0] &\in [LB_{1,m}(r_0, k_{1,m}), UB_{1,m}(r_0, k_{1,m})], \\ E[Y_0|R = r_1] &\in [LB_{0,m}(r_1, k_{0,m}), UB_{0,m}(r_1, k_{0,m})], \end{aligned}$$

where

$$\begin{aligned} LB_{1,m}(k_{1,m}, r_0) &= E[Y_1|R = r_1^*] - k_{1,m} \cdot |r_0 - r_1^*|, \\ UB_{1,m}(k_{1,m}, r_0) &= E[Y_1|R = r_1^*] + k_{1,m} \cdot |r_0 - r_1^*|, \\ LB_{0,m}(k_{0,m}, r_1) &= E[Y_0|R = r_0^*] - k_{0,m} \cdot |r_1 - r_0^*|, \\ UB_{0,m}(k_{0,m}, r_1) &= E[Y_0|R = r_0^*] + k_{0,m} \cdot |r_1 - r_0^*|. \end{aligned}$$

Moreover, these bounds are sharp.

Proof. We only prove the sharpness of the bounds with $LB_{1,m}$ and $UB_{1,m}$, and the sharpness of $LB_{0,m}$ and $UB_{0,m}$ can be shown by a similar way.

To verify the sharpness of $UB_{1,m}(k_{1,m}, r_0)$, consider the following DGP:

$$\tilde{g}_1(r) = \begin{cases} g_1(r) & \text{if } r \in \mathcal{R}_1, \\ \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y | R = \tilde{r}] + k_{1,m} \cdot |r - \tilde{r}| \right\} & \text{if } r \in \mathcal{R}_0. \end{cases}$$

This construction ensures that $\tilde{g}_1(r_0)$ attains the upper bound $UB_{1,m}(k_{1,m}, r_0)$ for every $r_0 \in \mathcal{R}_0$. Note that if $r \in \mathcal{R}_1$, then by construction, $\tilde{g}_1(r) = g_1(r)$, which is identified over $\mathcal{R}_1$. Thus Assumption C.1-(i) holds with this DGP. We now verify that this DGP satisfies Assumption C.1-(ii); that is, for any $r_1, r_2 \in \text{Supp}(R)$ such that $r_1 < r_2$, we have $|\tilde{g}_1(r_2) - \tilde{g}_1(r_1)| \leq k_{1,m}|r_2 - r_1|$. We consider the following three cases:

Case 1) $r_1, r_2 \in \mathcal{R}_1$: Since $\tilde{g}_1(r_j) = g_1(r_j)$ for $j = 1, 2$, and $g_1(\cdot)$ satisfies Assumption C.1-(ii), it is obvious that

$$\begin{aligned} |\tilde{g}_1(r_2) - \tilde{g}_1(r_1)| &= |g_1(r_2) - g_1(r_1)| \\ &\leq k_{1,m}|r_2 - r_1|. \end{aligned}$$

Case 2) $r_1 \in \mathcal{R}_0, r_2 \in \mathcal{R}_1$: We have

$$\begin{aligned}\tilde{g}_1(r_1) &= \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 \mid R = \tilde{r}] + k_{1,m} |r_1 - \tilde{r}| \right\} \\ &\leq E[Y_1 \mid R = r_2] + k_{1,m} |r_1 - r_2| \\ &= \tilde{g}_1(r_2) + k_{1,m} |r_1 - r_2|,\end{aligned}$$

and therefore,

$$\tilde{g}_1(r_1) - \tilde{g}_1(r_2) \leq k_{1,m} |r_1 - r_2|.$$

Conversely, since g_1 satisfies Assumption C.1-(ii), for all $\tilde{r} \in \mathcal{R}_1$,

$$g_1(r_2) \leq g_1(\tilde{r}) + k_1 |r_2 - \tilde{r}|,$$

so

$$\begin{aligned}\tilde{g}_1(r_2) &= g_1(r_2) \\ &\leq \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ g_1(\tilde{r}) + k_{1,m} |r_2 - \tilde{r}| \right\} \\ &\leq \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ g_1(\tilde{r}) + k_{1,m} |r_1 - \tilde{r}| \right\} + k_{1,m} |r_2 - r_1| \\ &= \tilde{g}_1(r_1) + k_{1,m} |r_1 - r_2|,\end{aligned}$$

which yields that

$$|\tilde{g}_1(r_2) - \tilde{g}_1(r_1)| \leq k_{1,m} |r_1 - r_2|.$$

Case 3): $r_1, r_2 \in \mathcal{R}_0$: By definition,

$$\tilde{g}_1(r_j) = \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 \mid R = \tilde{r}] + k_{1,m} |r_j - \tilde{r}| \right\}$$

for each $j = 1, 2$. Then, by the triangle inequality,

$$\begin{aligned}
& |\tilde{g}_1(r_2) - \tilde{g}_1(r_1)| \\
&= \left| \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] + k_{1,m} |r_2 - \tilde{r}| \right\} - \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] + k_{1,m} |r_1 - \tilde{r}| \right\} \right| \\
&\leq \left| \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] + k_{1,m} |r_1 - \tilde{r}| \right\} + k_{1,m} |r_2 - r_1| - \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] + k_{1,m} |r_1 - \tilde{r}| \right\} \right| \\
&= k_{1,m} |r_2 - r_1|.
\end{aligned}$$

In all, Assumption C.1-(ii) holds with the DGP.

We now consider the sharpness of the lower bound on $g_1(r_0)$. Consider the following DGP:

$$\bar{g}_1(r) = \begin{cases} g_1(r) & \text{if } r \in \mathcal{R}_1, \\ \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m} |r - \tilde{r}| \right\} & \text{if } r \in \mathcal{R}_0. \end{cases}$$

This construction ensures that $\bar{g}_1(r_0)$ attains the lower bound $LB_{1,m}(k_1, r_0)$ for every $r_0 \in \mathcal{R}_0$.

Note that if $r \in \mathcal{R}_1$, then by construction, $\bar{g}_1(r) = g_1(r)$, and thus Assumption C.1-(i) holds with this DGP.

We now verify that this DGP satisfies Assumption C.1-(ii); that is, for any $r_1 < r_2$, $|\bar{g}_1(r_2) - \bar{g}_1(r_1)| \leq k_{1,m} |r_2 - r_1|$. We consider the following three cases:

Case 1) $r_1, r_2 \in \mathcal{R}_1$: Since $\bar{g}_1(r_j) = g_1(r_j)$ for $j = 1, 2$ and $g_1(\cdot)$ satisfies Assumption C.1-(ii), it follows that

$$\begin{aligned}
|\bar{g}_1(r_2) - \bar{g}_1(r_1)| &= |g_1(r_2) - g_1(r_1)| \\
&\leq k_{1,m} |r_2 - r_1|.
\end{aligned}$$

Case 2) $r_1 \in \mathcal{R}_0$ and $r_2 \in \mathcal{R}_1$: By the definition of $\bar{g}_1$, we have

$$\begin{aligned}
\bar{g}_1(r_1) &= \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m} |r_1 - \tilde{r}| \right\} \\
&\geq E[Y_1 | R = r_2] - k_{1,m} |r_1 - r_2| \\
&= \bar{g}_1(r_2) - k_{1,m} |r_2 - r_1|,
\end{aligned}$$

and therefore,

$$\bar{g}_1(r_1) - \bar{g}_1(r_2) \geq -k_{1,m}|r_2 - r_1|.$$

Conversely, since g_1 satisfies Assumption C.1-(ii), it follows that, for all $\tilde{r} \in \mathcal{R}_1$,

$$g_1(\tilde{r}) \leq g_1(r_2) + k_{1,m}|r_2 - \tilde{r}|,$$

which is equivalent to $g_1(\tilde{r}) - k_{1,m}|r_2 - \tilde{r}| \leq g_1(r_2)$. Putting them together, we have

So

$$\begin{aligned} \bar{g}_1(r_2) &= g_1(r_2) \\ &\geq \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ g_1(\tilde{r}) - k_{1,m}|r_2 - \tilde{r}| \right\} \\ &\geq \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ g_1(\tilde{r}) - k_{1,m}|r_1 - \tilde{r}| \right\} - k_{1,m}|r_2 - r_1| \\ &= \bar{g}_1 - k_{1,m}|r_2 - r_1|. \end{aligned}$$

As a result, we obtain $|\bar{g}_1(r_2) - \bar{g}_1(r_1)| \leq k_{1,m}|r_2 - r_1|$ in this case.

Case 3) $r_1, r_2 \in \mathcal{R}_0$: By construction,

$$\bar{g}_1(r_j) = \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m}|r_j - \tilde{r}| \right\}$$

for each $j = 1, 2$. Then, by the triangle inequality, we have

$$\begin{aligned} &|\bar{g}_1(r_2) - \bar{g}_1(r_1)| \\ &= \left| \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m}|r_2 - \tilde{r}| \right\} - \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m}|r_1 - \tilde{r}| \right\} \right| \\ &\leq \left| \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m}|r_1 - \tilde{r}| \right\} + k_{1,m}|r_2 - r_1| - \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y_1 | R = \tilde{r}] - k_{1,m}|r_1 - \tilde{r}| \right\} \right| \\ &= k_{1,m}|r_2 - r_1|, \end{aligned}$$

implying that Assumption C.1-(ii) holds with this DGP. Therefore, the bounds $LB_{1,m}(k_{1,m}, r_0)$ and $UB_{1,m}(k_{1,m}, r_0)$ are sharp. We can show the sharpness of $LB_{0,m}(k_{0,m}, r_1)$ and $UB_{0,m}(k_{0,m}, r_1)$ by using a similar argument. ■

Theorem C.1 establishes two important results. First, it characterizes the identified set for the

conditional expectation function of the potential outcome on the running variable over regions in which point identification may fail. Second, it shows that the derived bounds are sharp. The results in Theorem C.1 is useful for constructing bounds on the unconditional means of Y_1 and Y_0 , and thus, allows us to construct bounds on the average treatment effect for the whole population. Specifically, suppose that we are interested in identification of $E[Y_1 - Y_0|R = r_1]$ for $r_1 \in \mathcal{R}_1 - \{c\}$. Note that $E[Y_1|R = r_1] = E[Y|R = r_1]$ is identified from the data. On the other hand, Theorem C.1 provides the sharp bounds on $E[Y_0|R = r_1]$, and the bounds are given by $[LB_{0,m}(k_{0,m}, r_1), UB_{0,m}(k_{0,m}, r_1)]$. As a result, it is straightforward to see that

$$E[Y_1 - Y_0|R = r_1] \in [E[Y_1|R = r_1] - UB_{0,m}(k_{0,m}, r_1), E[Y_1|R = r_1] - LB_{0,m}(k_{0,m}, r_1)].$$

The same argument can be applied pointwise to each value of the running variable in its support. Integrating the resulting bounds with respect to the distribution of the running variable then yields bounds on the average treatment effect. The following corollary formalizes this result.

Corollary C.2. *Suppose that Assumption C.1 holds and that $\mathcal{R}_1 \cup \mathcal{R}_0 = \text{Supp}(R)$ with $\Pr(\mathcal{R}_1 \cap \mathcal{R}_0) = 0$. Let $\mathcal{C}$ be a measurable subset of $\text{Supp}(R)$ and $ATE(\mathcal{C}) \equiv E[Y_1 - Y_0|R \in \mathcal{C}]$. Then, we have*

$$ATE(\mathcal{C}) \in [LB^{ATE}(k_{0,m}, k_{1,m}; \mathcal{C}), UB^{ATE}(k_{0,m}, k_{1,m}; \mathcal{C})],$$

where

$$\begin{aligned} LB^{ATE}(k_{0,m}, k_{1,m}; \mathcal{C}) &\equiv \int_{\mathcal{R}_0 \cap \mathcal{C}} LB_{1,m}(k_{1,m}, r) - E[Y|R = r] dF_R(r) \\ &\quad + \int_{\mathcal{R}_1 \cap \mathcal{C}} E[Y|R = r] - UB_{0,m}(k_{0,m}, r) dF_R(r), \\ UB^{ATE}(k_{0,m}, k_{1,m}; \mathcal{C}) &\equiv \int_{\mathcal{R}_1 \cap \mathcal{C}} E[Y|R = r] - LB_{0,m}(k_{0,m}, r) dF_R(r) \\ &\quad + \int_{\mathcal{R}_0 \cap \mathcal{C}} UB_{1,m}(k_{1,m}, r) - E[Y|R = r] dF_R(r). \end{aligned}$$

When $\mathcal{C} = \text{Supp}(R)$, then the resulting bounds are for the average treatment effect for the whole population. We note that the bounds in Corollary C.2 need not be sharp. To obtain sharp bounds for certain causal parameters, we instead rely on identification of the distributions of the potential outcomes.

The following assumption can be considered as a version of SMRV assumption (Assumption 3.3) for the counterfactual mean functions, which we call the “monotone running variable” assumption.

Assumption C.2 (Monotone Running Variable (MRV)). *For any $r, \tilde{r} \in \text{Supp}(R)$ such that $r \geq \tilde{r}$, we have*

$$E[Y_d|R = r] \geq E[Y_d|R = \tilde{r}]$$

for each $d \in \{0, 1\}$.

Assumption C.2 posits that the average potential outcome associated with a higher value of the running variable is not smaller than that associated with a lower value. This assumption is closely related to the MIV assumption in Manski and Pepper (2000) and Manski and Pepper (2009).

Under Assumptions C.1 and C.2, we obtain the following bounds on the conditional expectation functions of the potential outcomes:

$$\begin{aligned} E[Y_1|R = r_0] &\in [LB_1^{MRV}(k_{1,m}, r_0), UB_1^{MRV}(k_{1,m}, r_0)], \\ E[Y_0|R = r_1] &\in [LB_0^{MRV}(k_{0,m}, r_1), UB_0^{MRV}(k_{0,m}, r_1)], \end{aligned}$$

where

$$\begin{aligned} LB_1^{MRV}(k_{1,m}, r_0) &\equiv E[Y_1|R = r_1^*] - k_{1,m} \cdot |r_0 - r_1^*|, \\ UB_1^{MRV}(k_{1,m}, r_0) &\equiv E[Y_1|R = r_1^*], \\ LB_0^{MRV}(k_{0,m}, r_1) &\equiv E[Y_0|R = r_0^*], \\ UB_0^{MRV}(k_{0,m}, r_1) &\equiv E[Y_0|R = r_0^*] + k_{0,m} \cdot |r_0^* - r_1|, \end{aligned}$$

where $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$.

The above bounds are similar to those in Theorem C.1. However, $UB_1^{MRV}(k_{1,m}, r_0)$ and $LB_0^{MRV}(k_{0,m}, r_1)$ are tighter than their respective counterparts. This tightening follows from the fact that Assumption C.2 imposes a one-sided monotonicity restriction on the conditional mean function of the potential outcomes with respect to the running variable, which is consistent with the results established in the literature.